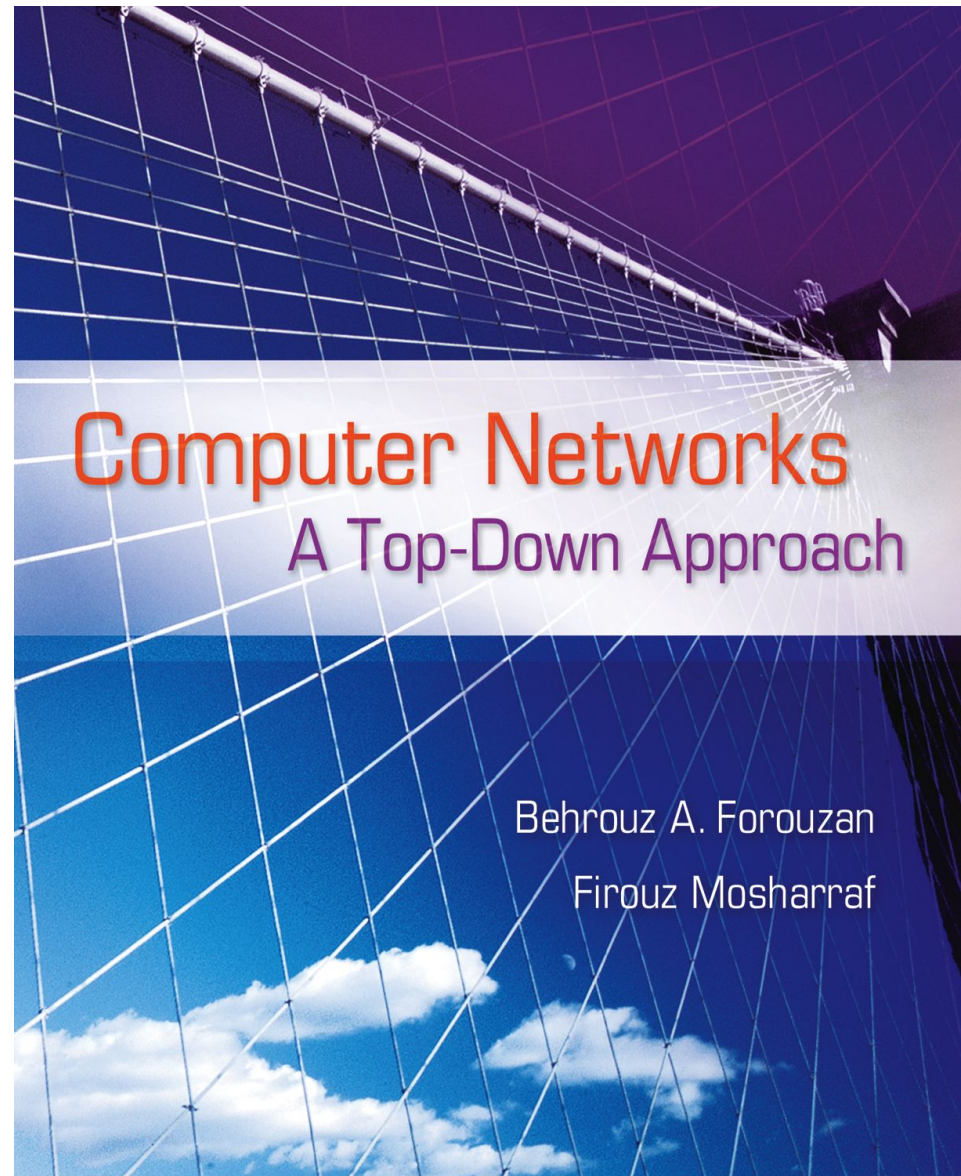
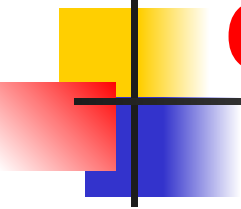


*Chapter 6*

*Wireless  
Networks  
and  
Mobile IP*





# Chapter6: Outline

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## **6.1** *WIRLESS LANS*

## **6.2** *OTHER WIRELESS NETWORKS*

## **6.3** *MOBILE IP*

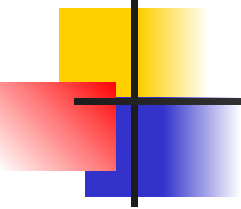


# Chapter 6: Objective

- *We introduce wired LANs, using IEEE project 802.11, the dominant standard. Next, we cover the Bluetooth LANs that are used as stand-alone LANs with many applications. We also discuss WiMAX technology, which is the counterpart of last-mile wired networks such as DSL or cable.*
- *We then discuss other wireless networks that can be categorized as wireless WANs or wireless broadband networks. For this purpose, we first discuss the channelization access method that is used in cellular telephones.*
- *We finally talk about mobile IP, which provides mobile access to the Internet. Our discussion include addressing, a big issue in mobile networking, and three phases of mobile access.*

## 6-1 WIRELESS LANS

*Wireless communication is one of the fastest-growing technologies. The demand for connecting devices without the use of cables is increasing everywhere. Wireless LANs can be found on college campuses, in office buildings, and in many public areas.*



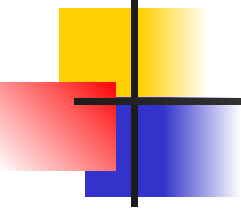
## 6.1.1 Introduction

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*Before we discuss a specific protocol related to wireless LANs, let us talk about them in general.*

### *Architectural Comparison*

- ◆ *Medium*
- ◆ *Hosts*
- ◆ *Isolated LANs*
- ◆ *Connection to Other Networks*
- ◆ *Moving between Environments*



## 6.1.1 (continued)

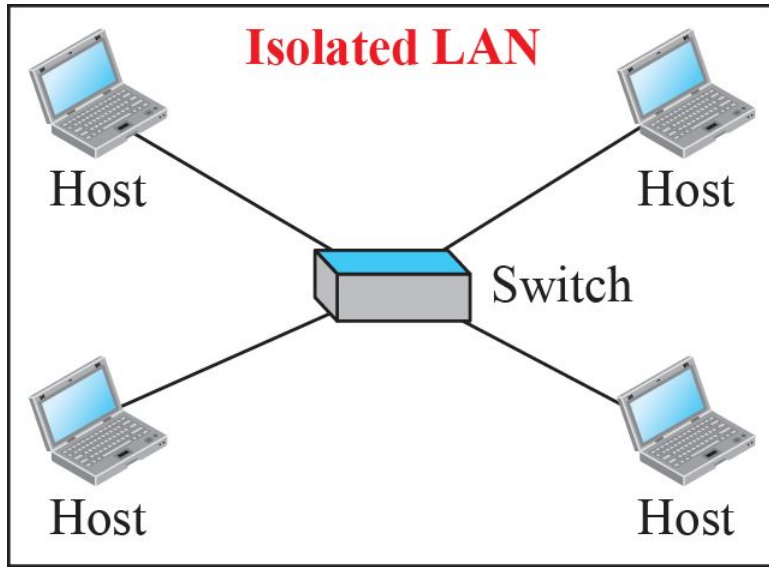
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### *Characteristics*

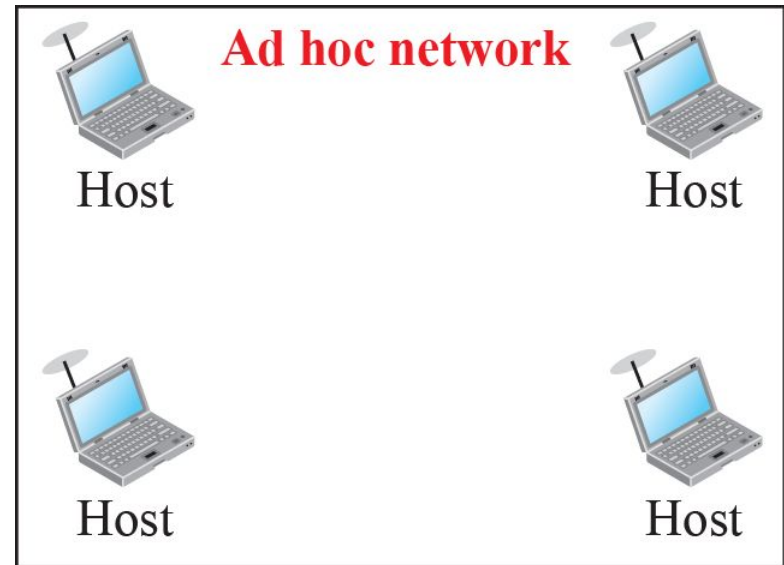
- ◆ *Attenuation*
- ◆ *Interference*
- ◆ *Multipath Propagation*
- ◆ *Error*

### *Access Control*

**Figure 6.1:** *Isolated LANs: wired versus wireless*

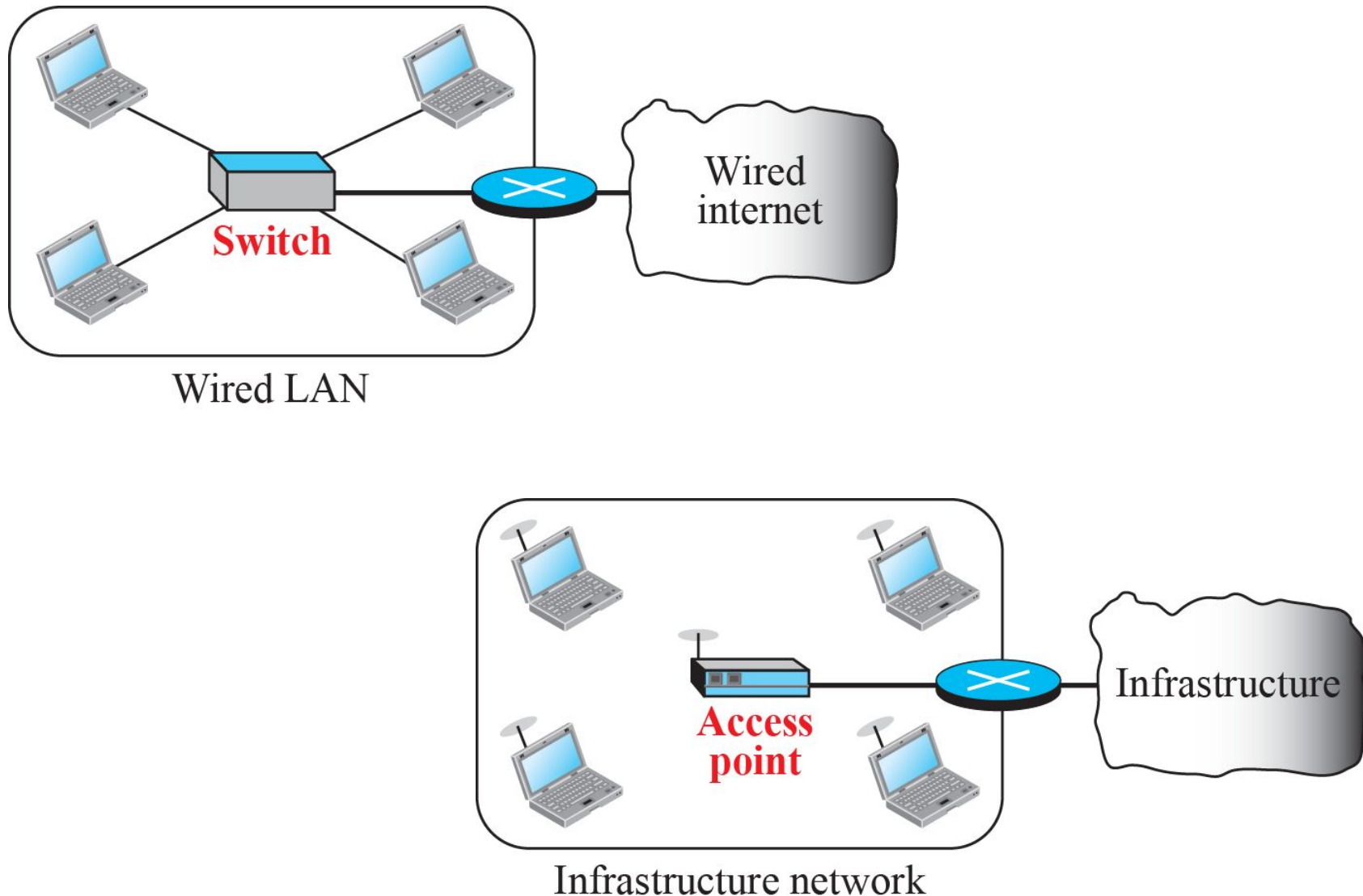


Wired



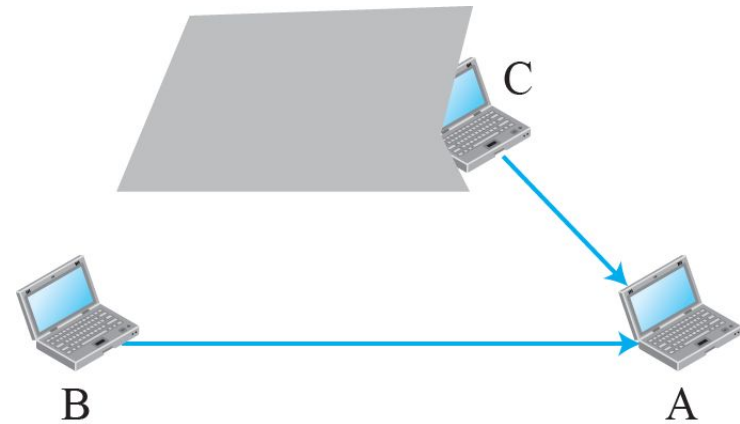
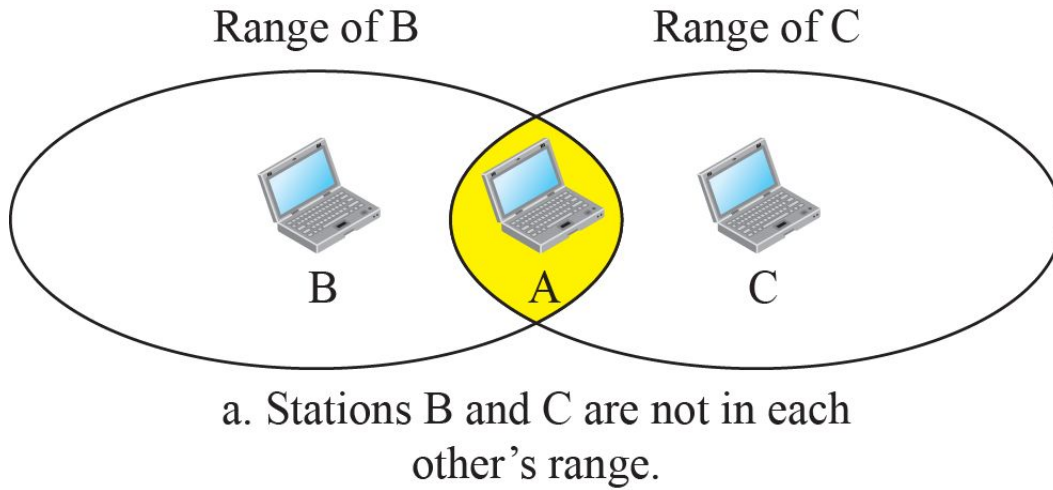
Wireless

**Figure 6.2:** *Connection of a wired LAN and a wireless LAN to other networks*





**Figure 6.3:** *Hidden station problem*



b. Stations B and C are hidden from each other.



## 6.1.2 IEEE 802.11 Project

*IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers. In some countries, including the United States, the public uses the term WiFi (short for wireless fidelity) as a synonym for wireless LAN. WiFi, however, is a wireless LAN that is certified by the WiFi Alliance, a global, nonprofit industry association of more than 300 member companies devoted to promoting the growth of wireless LANs.*

### □ *Architecture*

- ◆ *Basic Service Set*
- ◆ *Extended Service Set*
- ◆ *Station Types*

### □ *MAC Sublayer*

- ◆ *Distributed Coordination Function (DCF)*
- ◆ *Point Coordination Function (PCF)*
- ◆ *Fragmentation*
- ◆ *Frame Format*
- ◆ *Frame Types*



## 6.1.2 (continued)

---

□ *Addressing Mechanism*

□ *Exposed Station Problem*

□ *Physical Layer*

◆ *IEEE 802.11 FHSS*

◆ *IEEE 802.11 DSSS*

◆ *IEEE 802.11 Infrared*

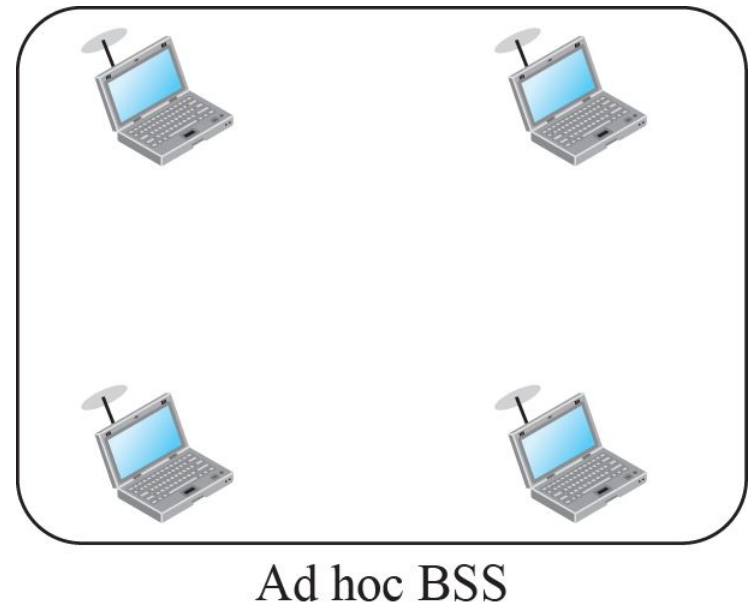
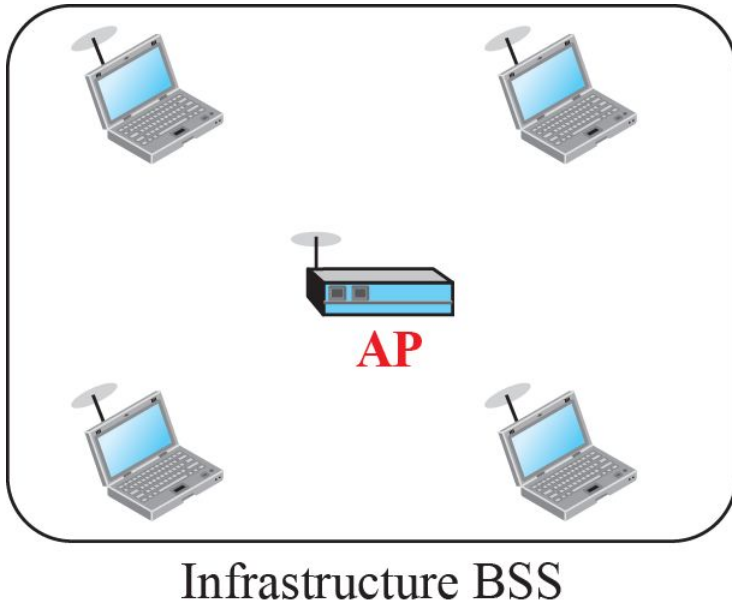
◆ *IEEE 802.11a OFDM*

◆ *IEEE 802.11b DSSS*

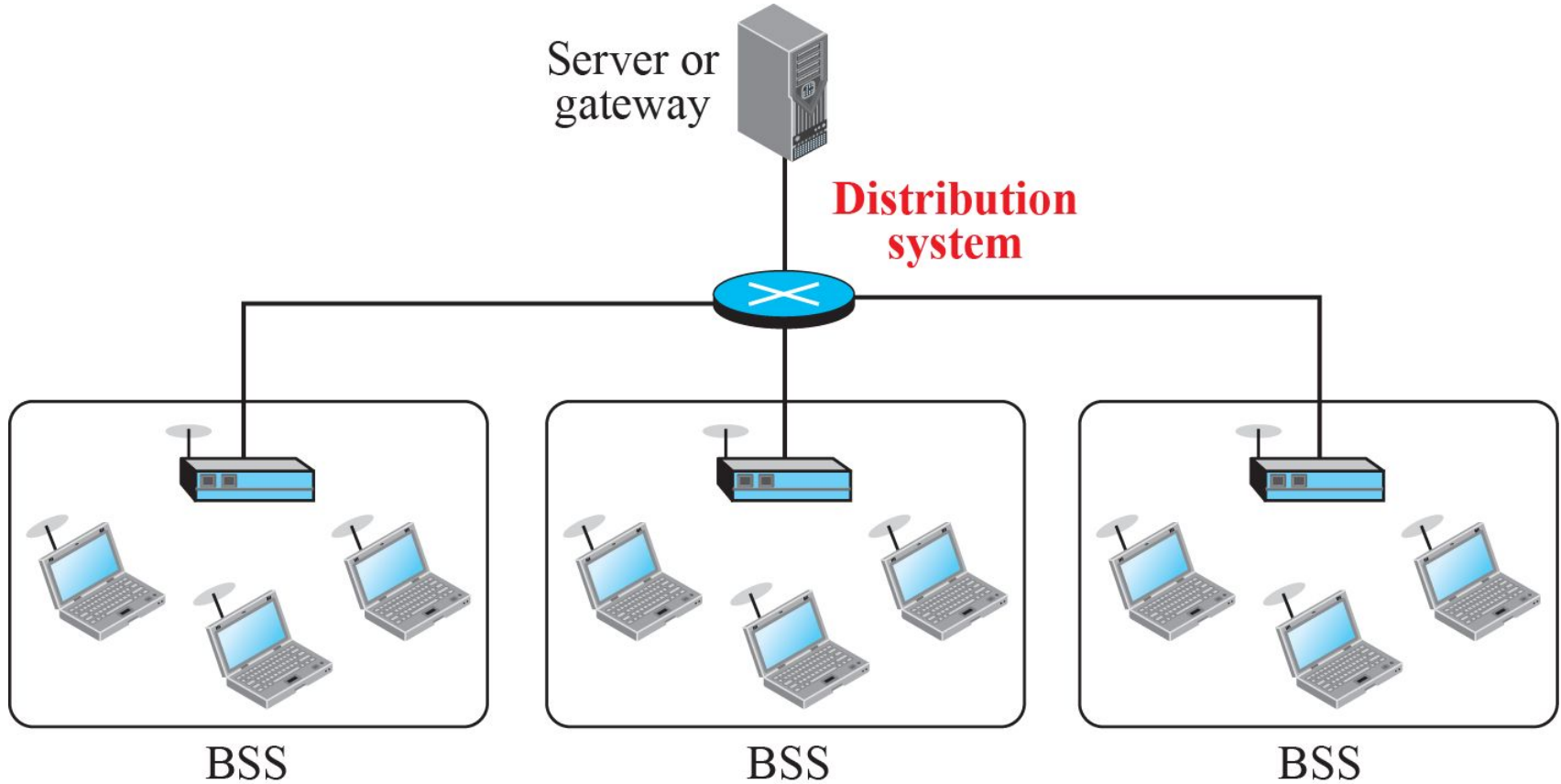
◆ *IEEE 802.11g*

◆ *IEEE 802.11n*

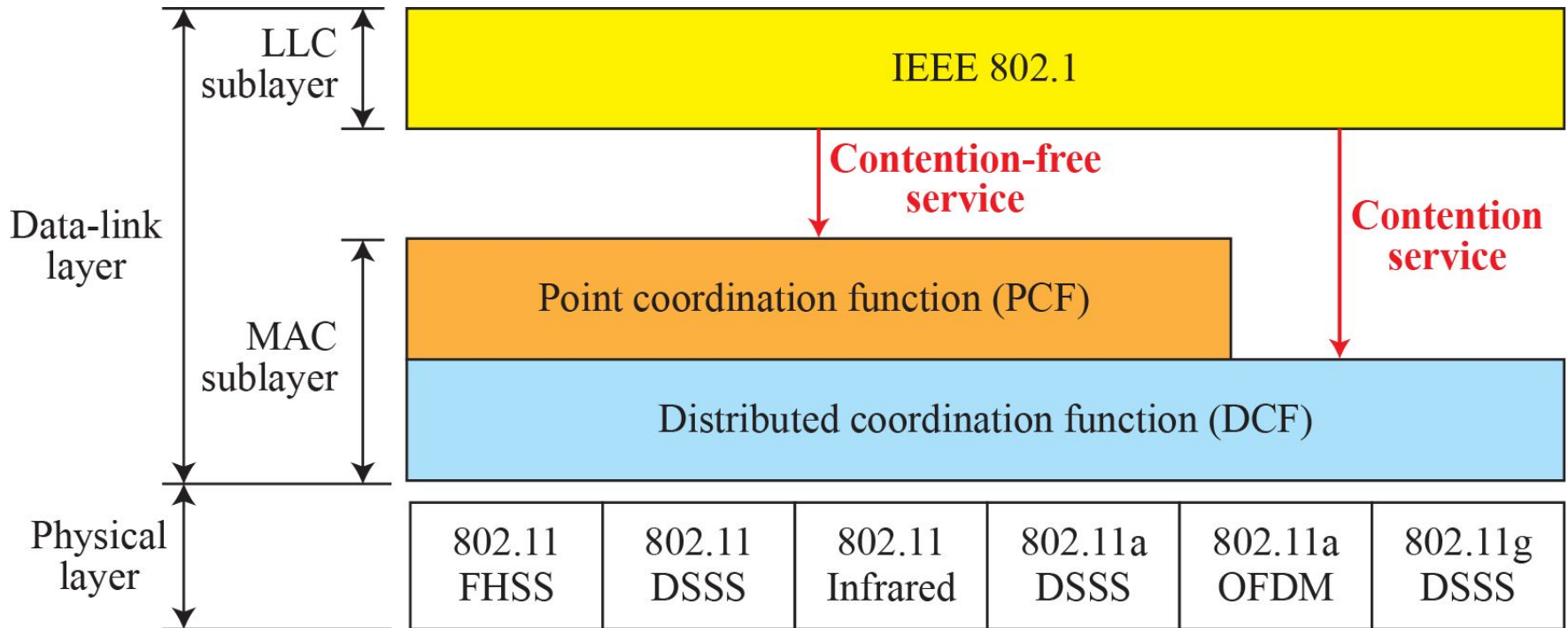
**Figure 6.4:** *Basic service sets (BSSs)*



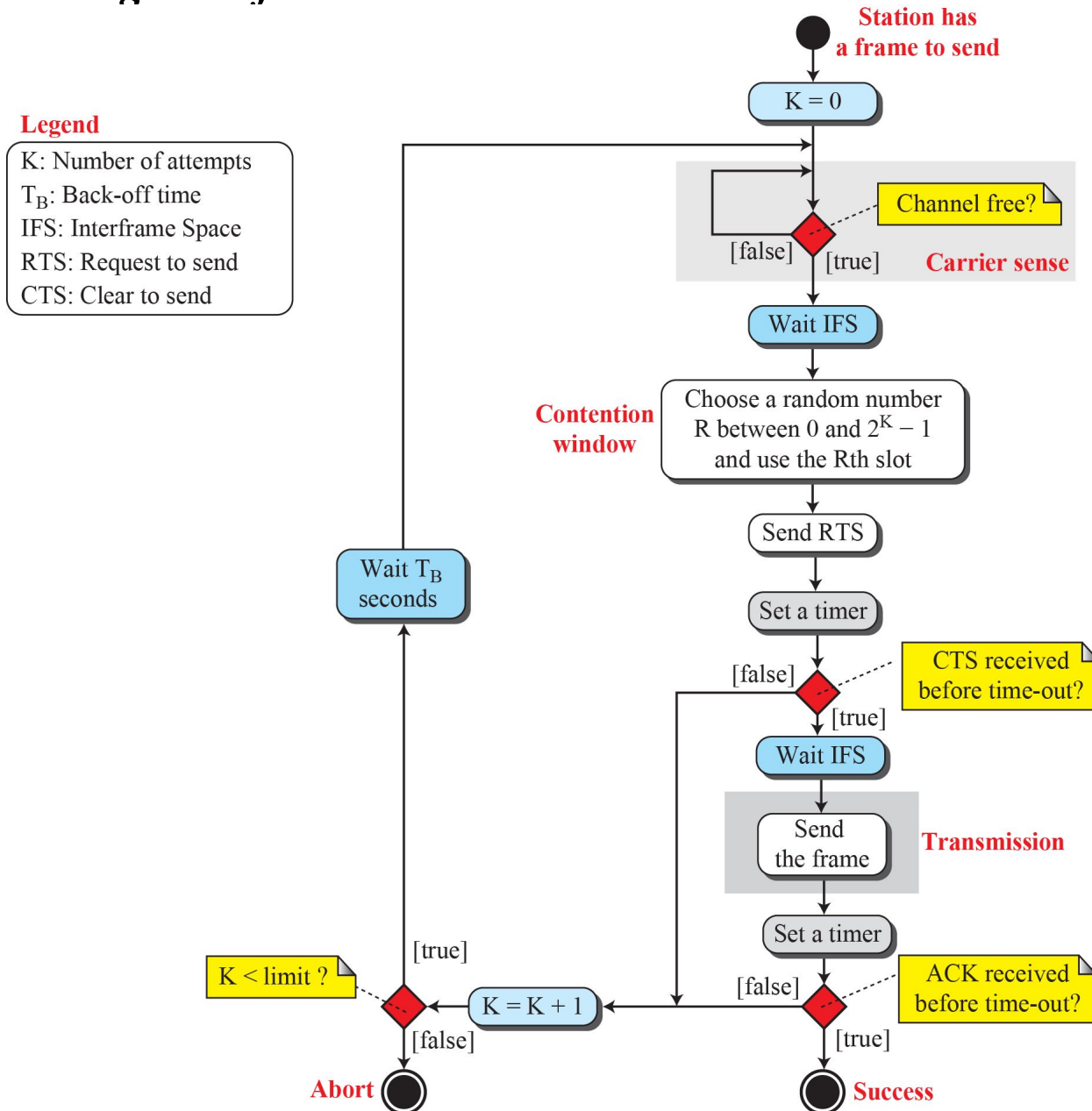
**Figure 6.5:** *Extended service set (ESS)*



**Figure 6.6:** *MAC layers in IEEE 802.11 standard*

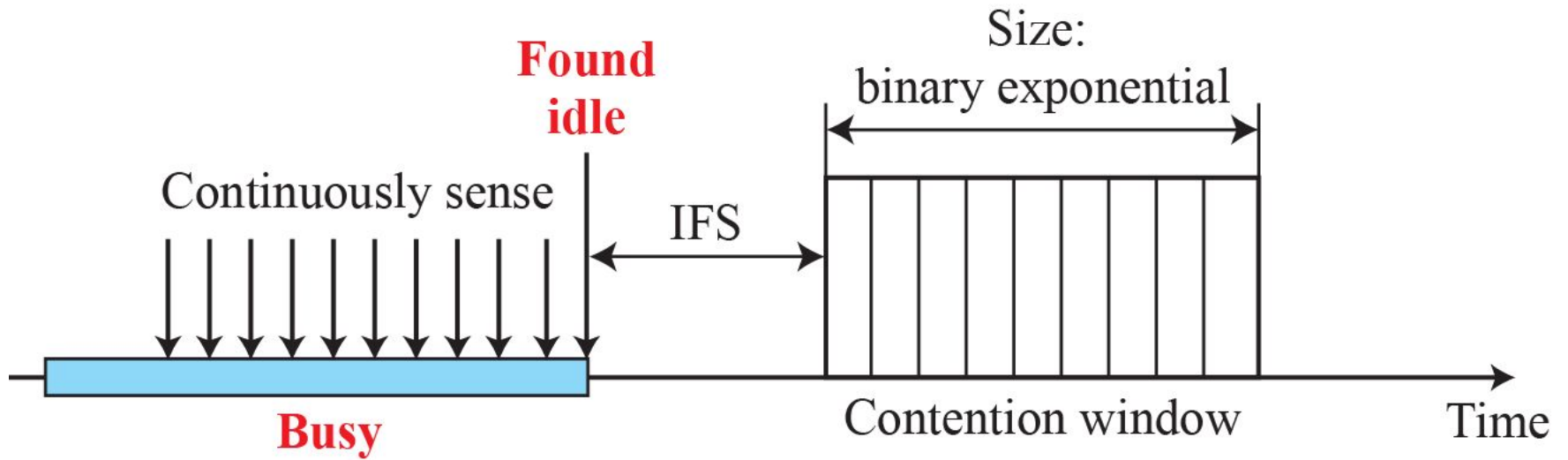


**Figure 6.7: Flow diagram of CSMA/CA**

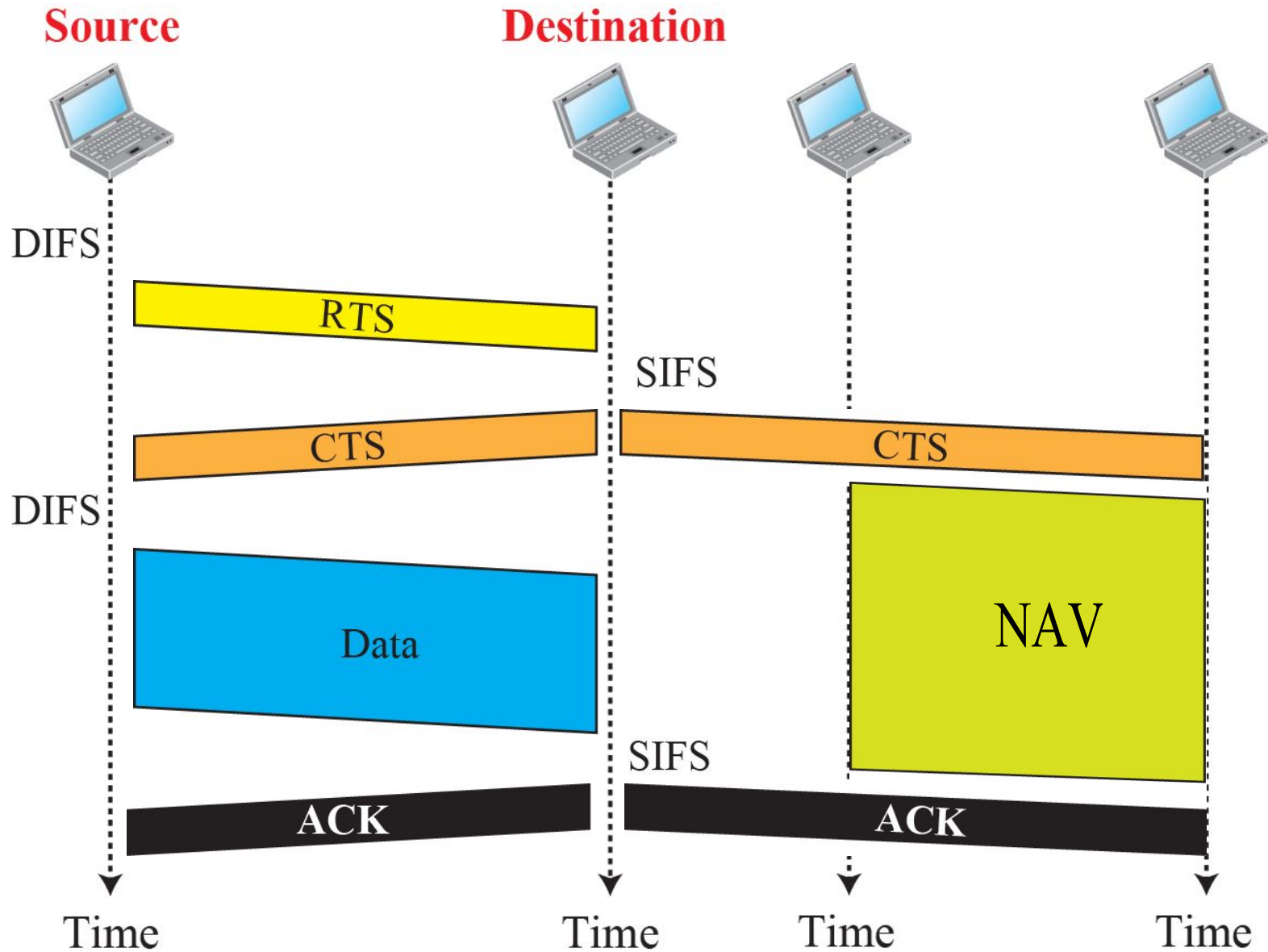




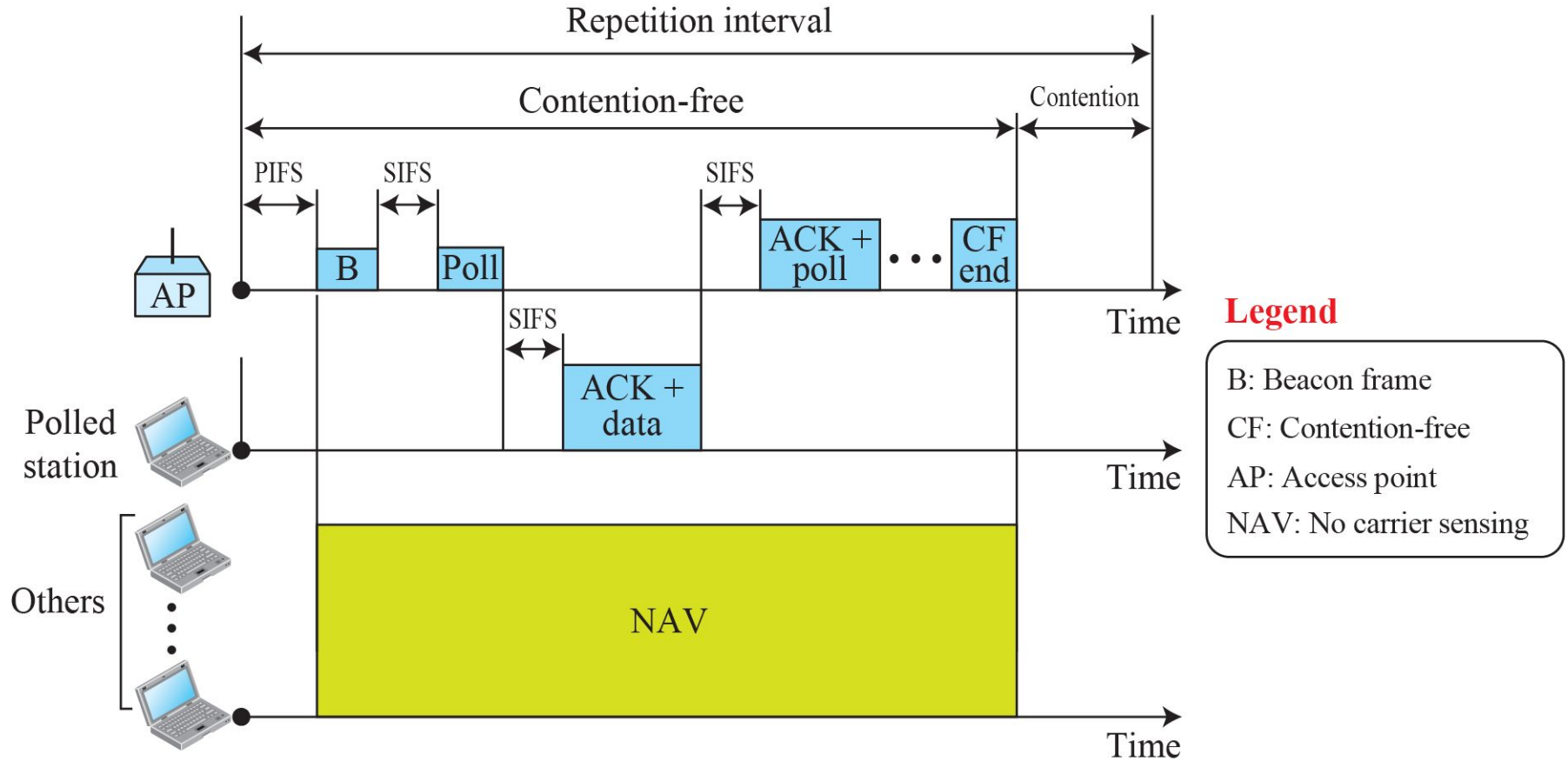
**Figure 6.8:** *Contention window*



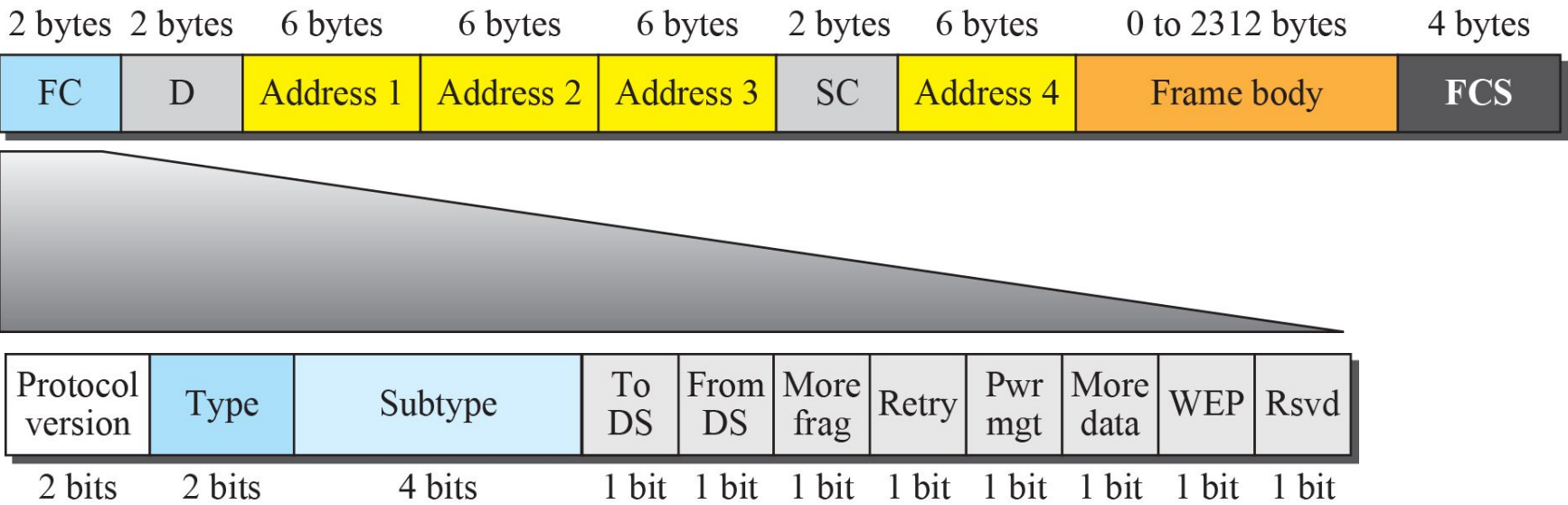
**Figure 6.9:** CSMA/CA and NAV



**Figure 6.10:** *Example of repetition interval*



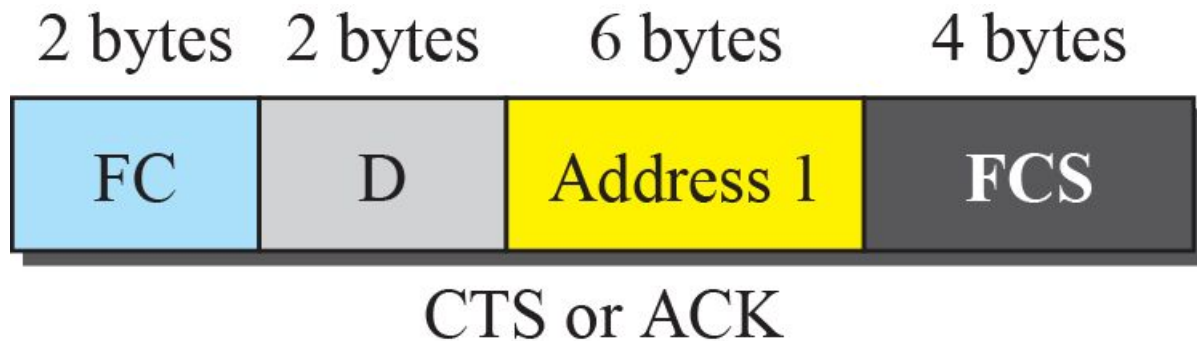
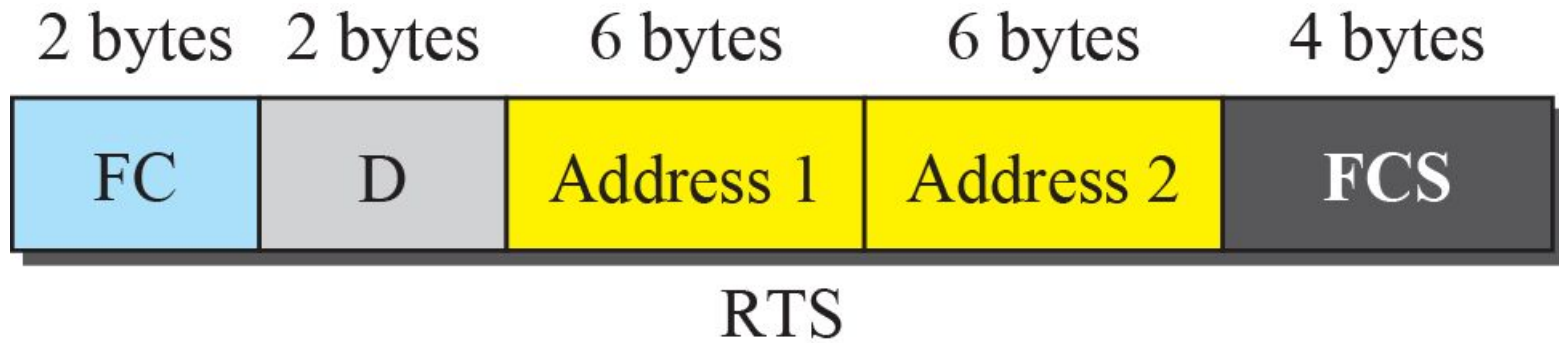
**Figure 6.11:** *Frame format*



**Table 6.1: Subfields in FC field**

| <i>Field</i> | <i>Explanation</i>                                               |
|--------------|------------------------------------------------------------------|
| Version      | Current version is 0                                             |
| Type         | Type of information: management (00), control (01), or data (10) |
| Subtype      | Subtype of each type (see Table 6.2)                             |
| To DS        | Defined later                                                    |
| From DS      | Defined later                                                    |
| More flag    | When set to 1, means more fragments                              |
| Retry        | When set to 1, means retransmitted frame                         |
| Pwr mgt      | When set to 1, means station is in power management mode         |
| More data    | When set to 1, means station has more data to send               |
| WEP          | Wired equivalent privacy (encryption implemented)                |
| Rsvd         | Reserved                                                         |

**Figure 6.12: Control frames**





**Table 6.2:** Values of subfields in control frames

| <i>Subtype</i> | <i>Meaning</i>        |
|----------------|-----------------------|
| 1011           | Request to send (RTS) |
| 1100           | Clear to send (CTS)   |
| 1101           | Acknowledgment (ACK)  |

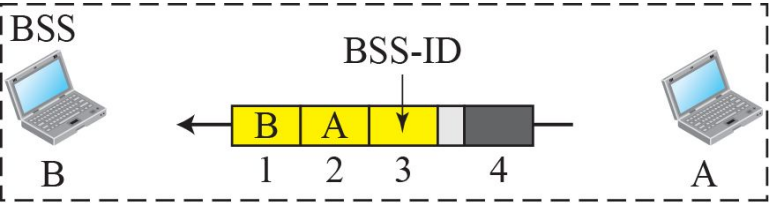


## Table 6.3: Addresses

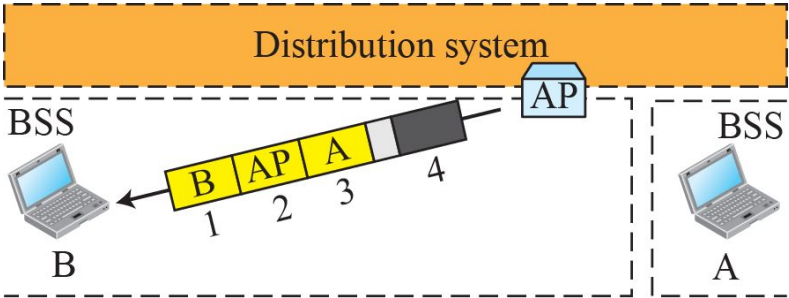
| <i>To<br/>DS</i> | <i>From<br/>DS</i> | <i>Address<br/>1</i> | <i>Address<br/>2</i> | <i>Address<br/>3</i> | <i>Address<br/>4</i> |
|------------------|--------------------|----------------------|----------------------|----------------------|----------------------|
| 0                | 0                  | Destination          | Source               | BSS ID               | N/A                  |
| 0                | 1                  | Destination          | Sending AP           | Source               | N/A                  |
| 1                | 0                  | Receiving AP         | Source               | Destination          | N/A                  |
| 1                | 1                  | Receiving AP         | Sending AP           | Destination          | Source               |



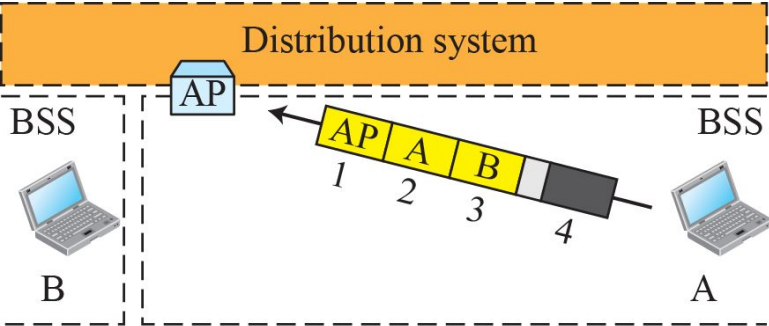
**Figure 6.13:** *Addressing mechanisms*



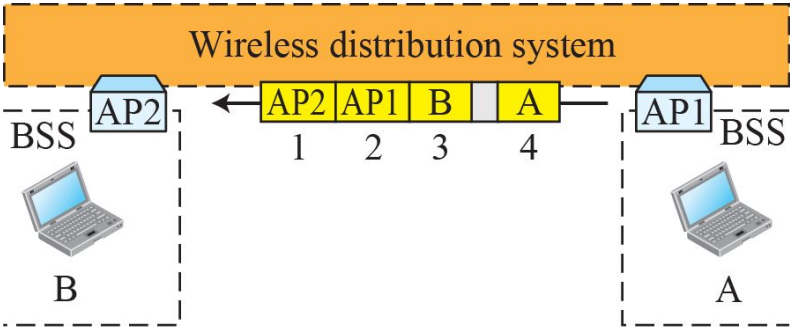
a. Case 1



b. Case 2

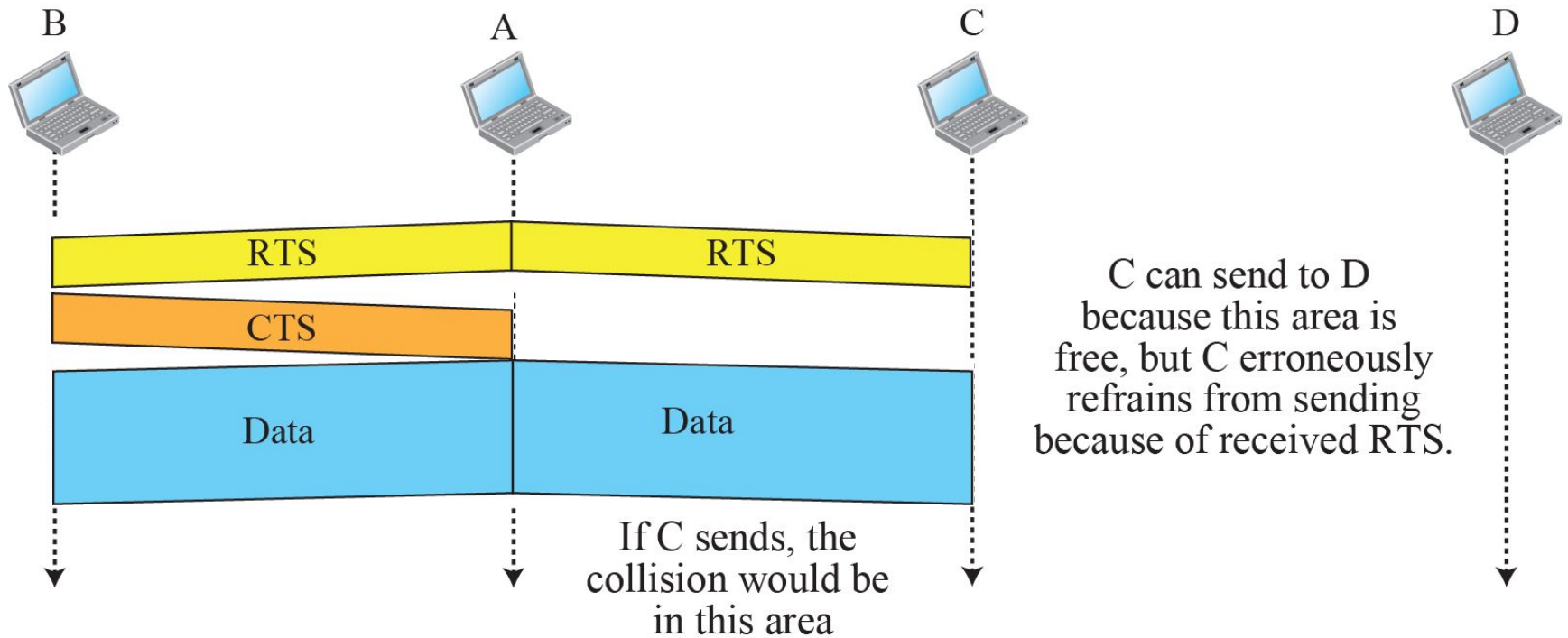


c. Case 3



d. Case 4

**Figure 6.14:** *Exposed station problem*

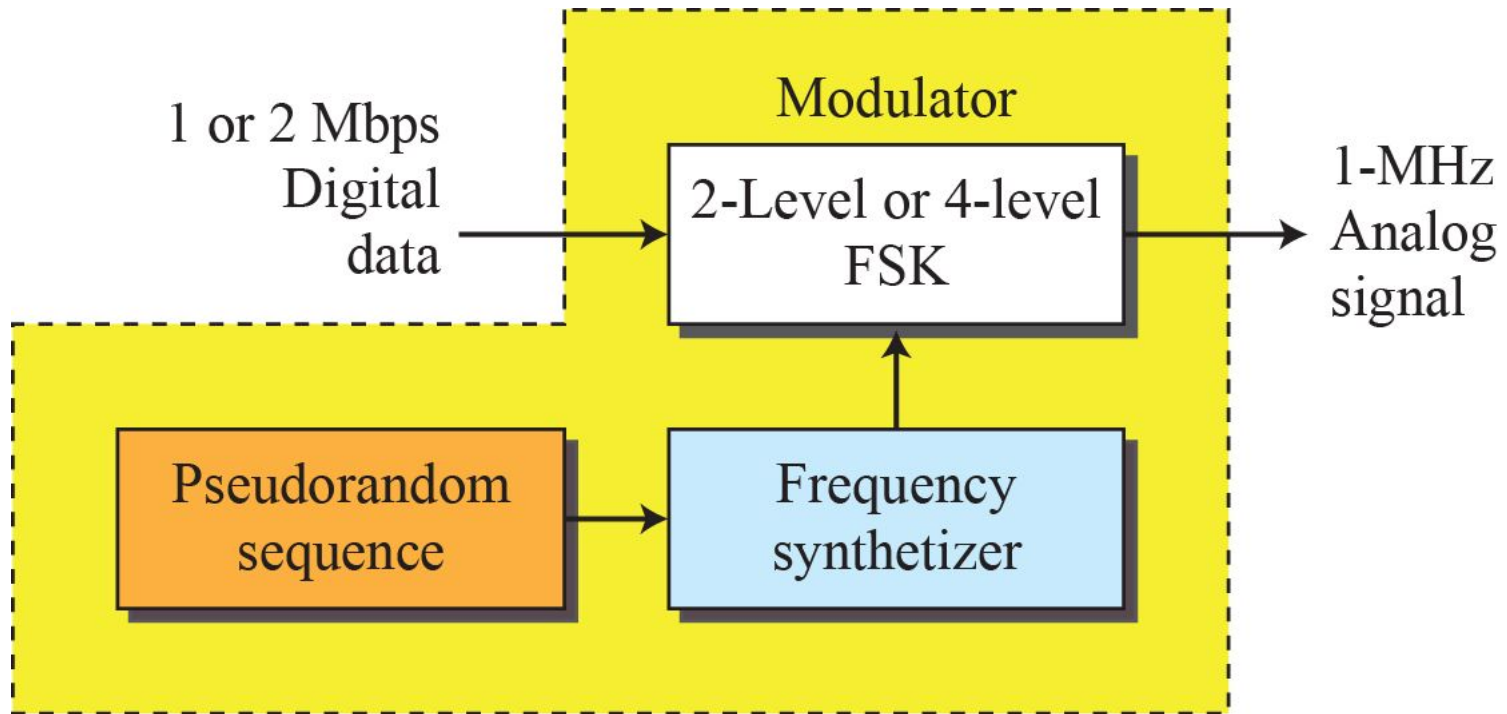




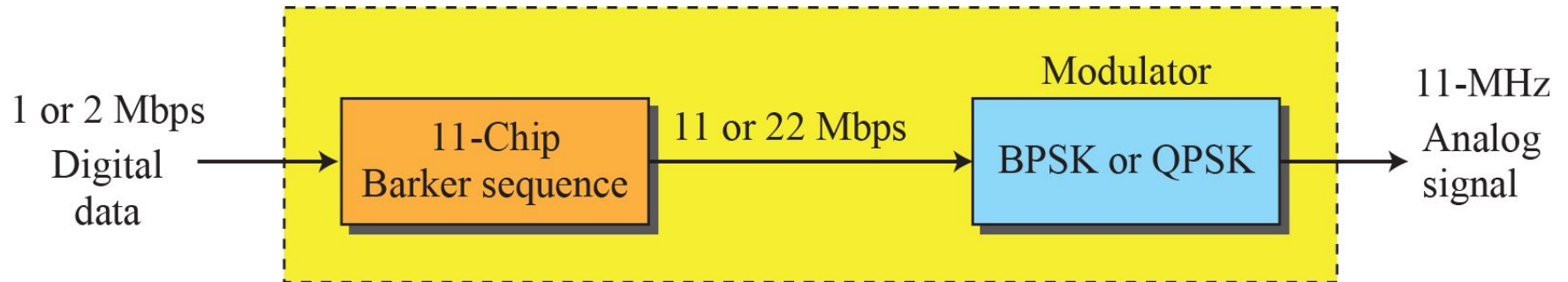
**Table 6.4: Specifications**

| <i>IEEE</i> | <i>Technique</i> | <i>Band</i>     | <i>Modulation</i> | <i>Rate (Mbps)</i> |
|-------------|------------------|-----------------|-------------------|--------------------|
| 802.11      | FHSS             | 2.400–4.835 GHz | FSK               | 1 and 2            |
|             | DSSS             | 2.400–4.835 GHz | PSK               | 1 and 2            |
|             | None             | Infrared        | PPM               | 1 and 2            |
| 802.11a     | OFDM             | 5.725–5.850 GHz | PSK or QAM        | 6 to 54            |
| 802.11b     | DSSS             | 2.400–4.835 GHz | PSK               | 5.5 and 11         |
| 802.11g     | OFDM             | 2.400–4.835 GHz | Different         | 22 and 54          |
| 802.11n     | OFDM             | 5.725–5.850 GHz | Different         | 600                |

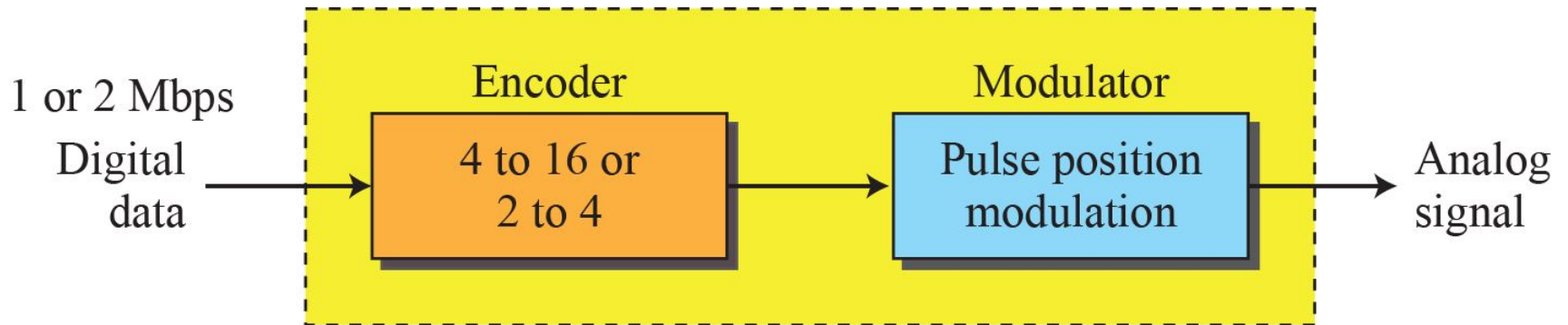
**Figure 6.15:** *Physical layer of IEEE 802.11 FHSS*



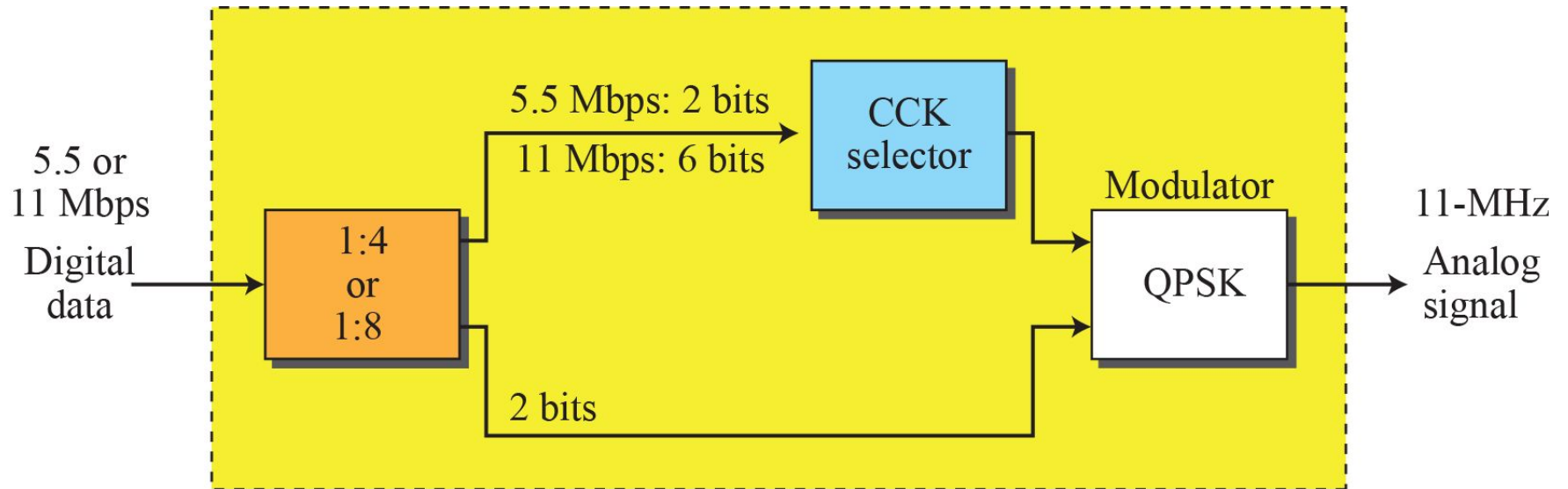
**Figure 6.16:** *Physical layer of IEEE 802.11 DSSS*



**Figure 6.17:** *Physical layer of IEEE 802.11 infrared*



**Figure 6.18:** *Physical layer of IEEE 802.11b*





## 6.1.3 Bluetooth

---

*Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, printers, and even coffee makers when they are at a short distance from each other. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously; the devices, sometimes called gadgets, find each other and make a network called a piconet.*



### *Architecture*

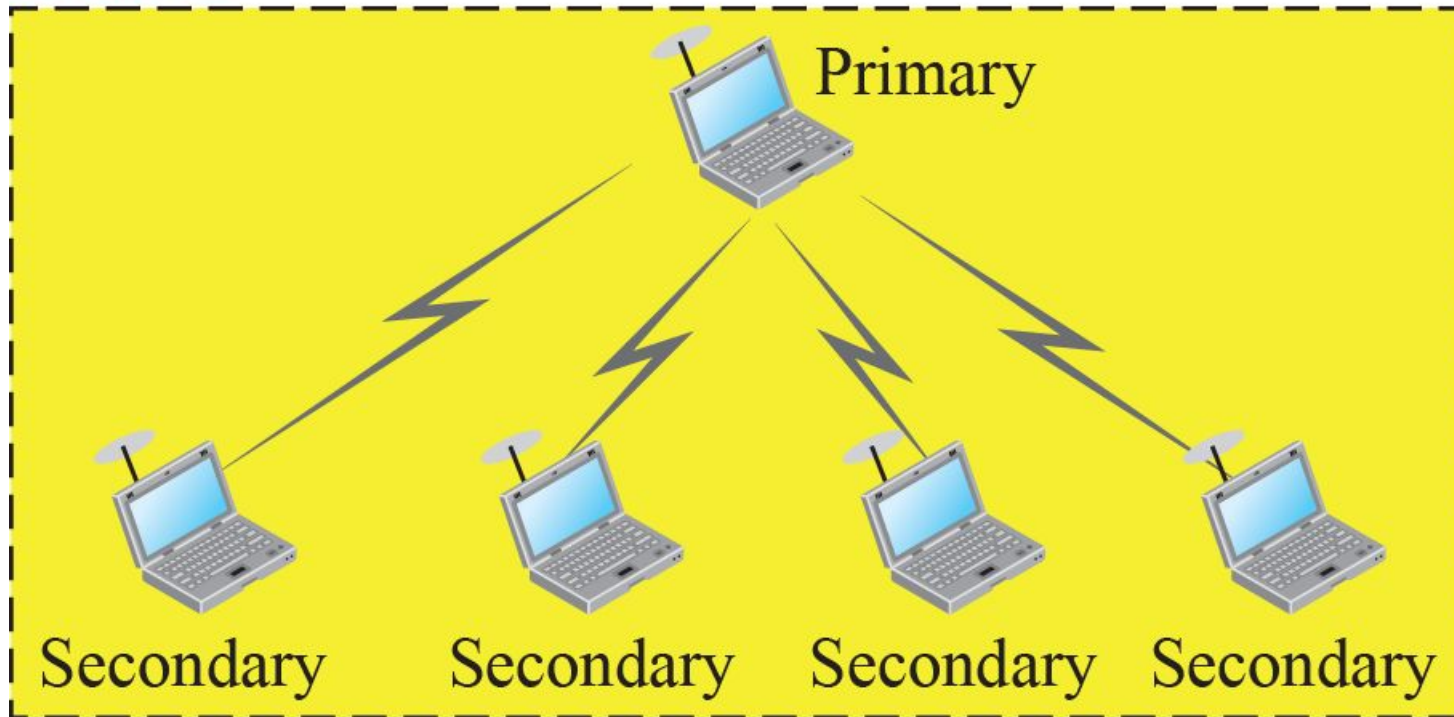
- ◆ *Piconets*
- ◆ *Scatternet*
- ◆ *Bluetooth Devices*

### *Bluetooth Layers*

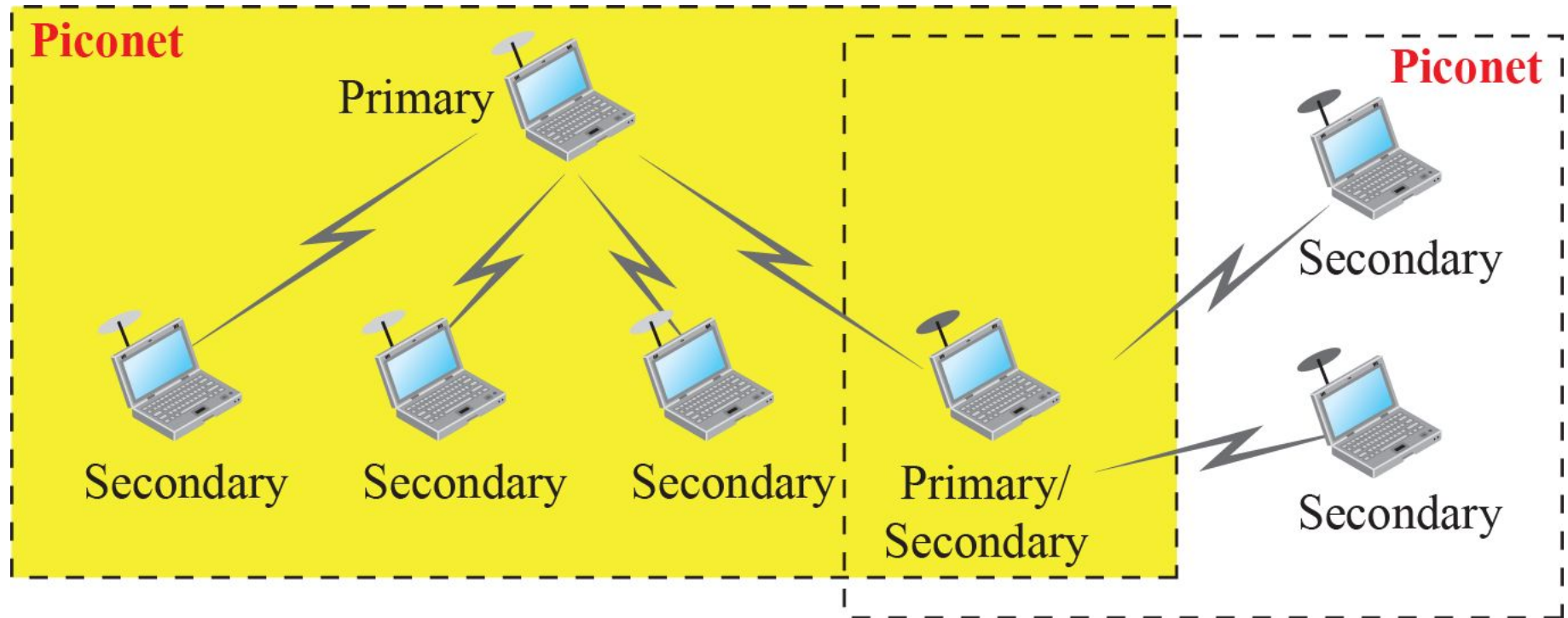
- ◆ *L2CAP*
- ◆ *Baseband Layer*
- ◆ *Radio Layer*

**Figure 6.19:** *Piconet*

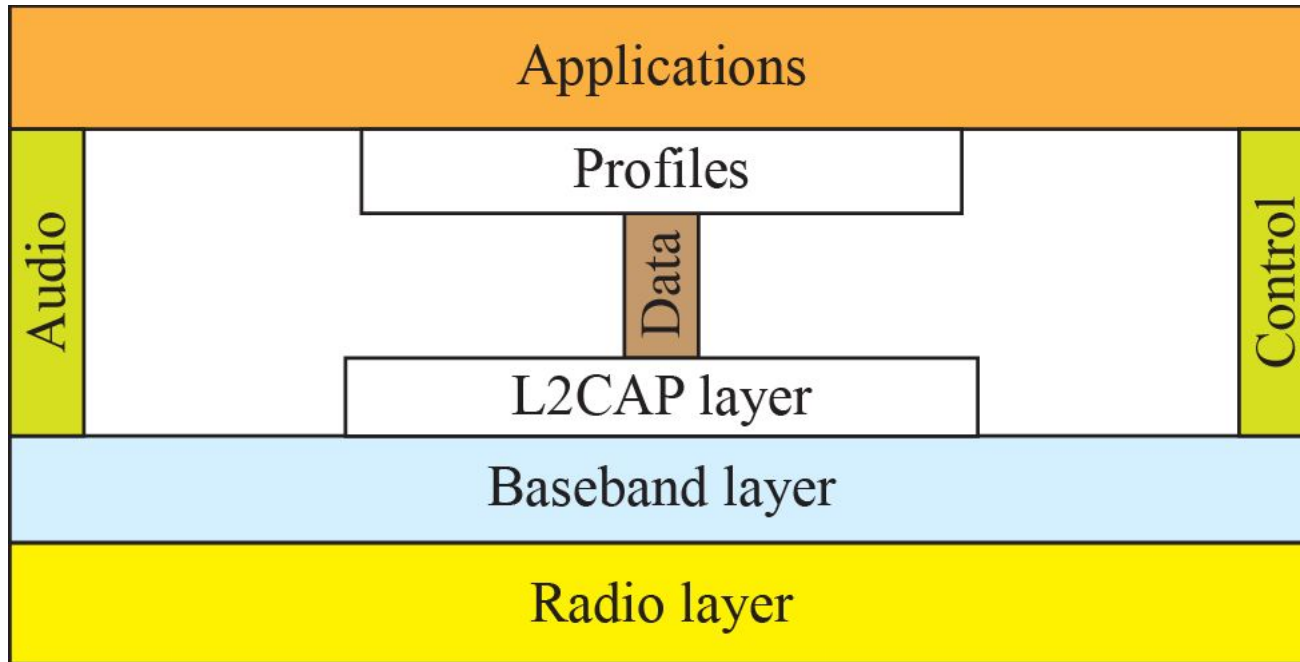
## Piconet



**Figure 6.20: Scatternet**



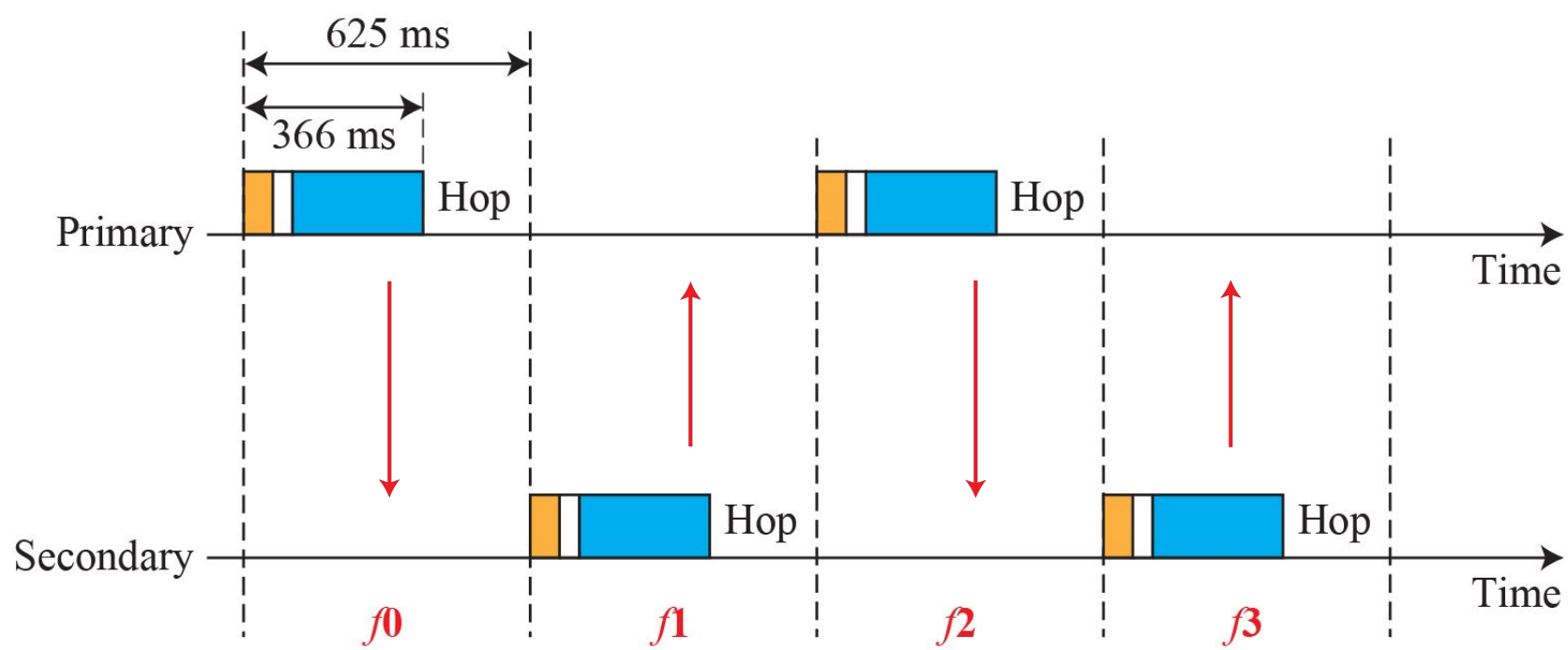
**Figure 6.21:** *Bluetooth layers*



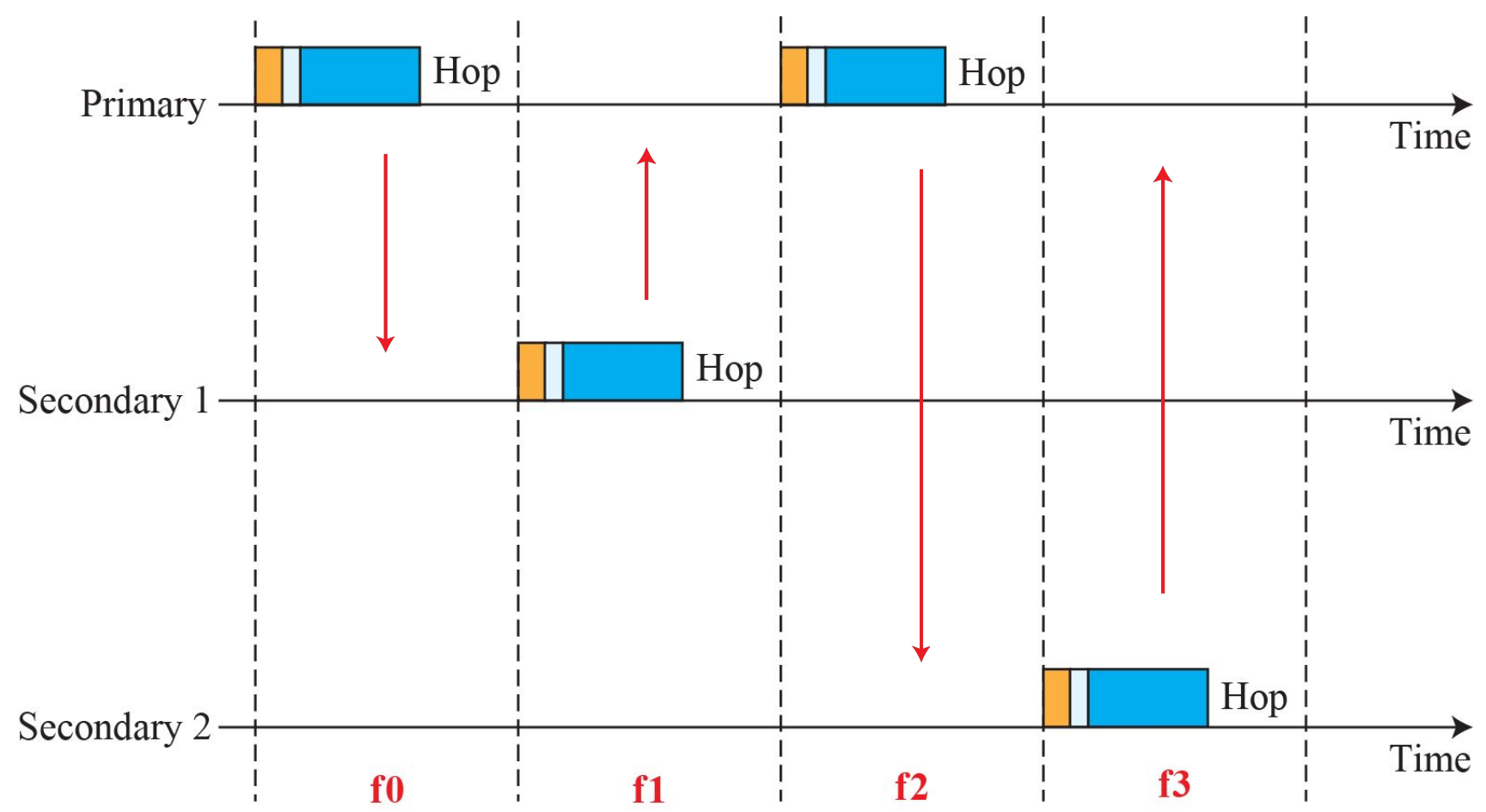
**Figure 6.22:** *L2CAP data packet format*



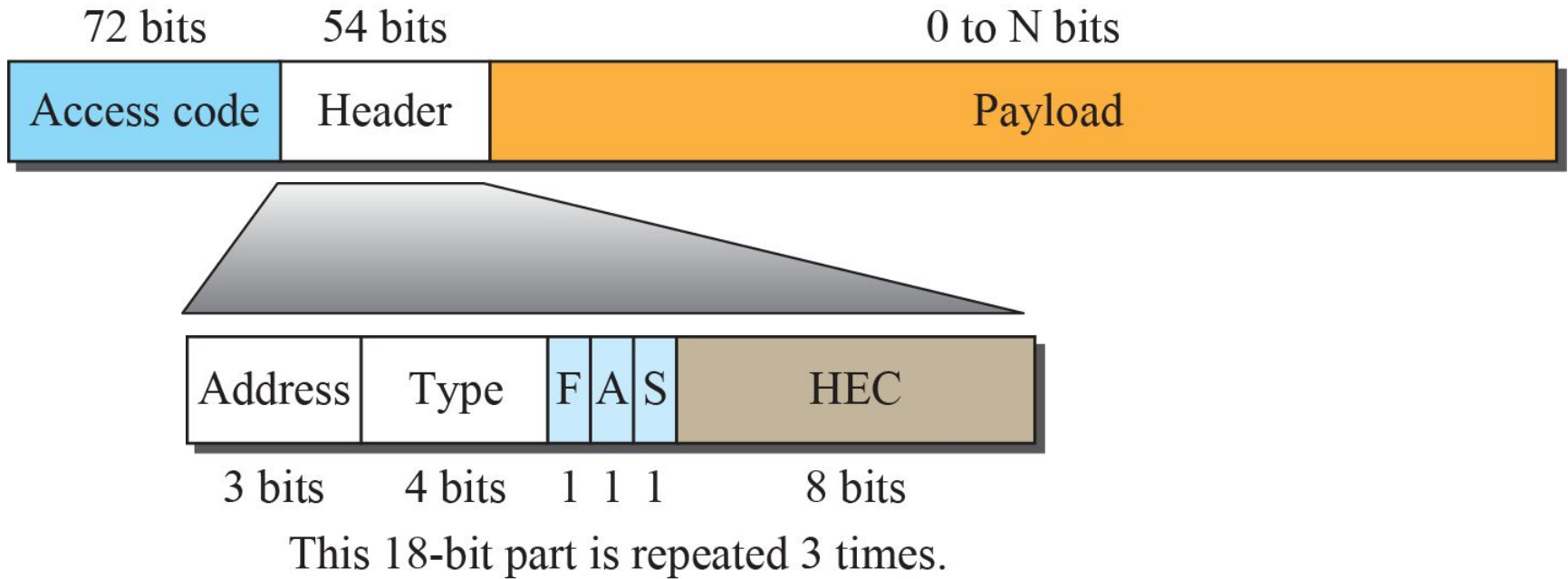
**Figure 6.23:** *Single-secondary communication*



**Figure 6.24:** *Multiple-secondary communication*



**Figure 6.25: Frame format types**



**N = 240 for 1-slot frame**

**N = 1490 for 3-slot frame**

**N = 2740 for 5-slot frame**





## 6.1.4 WiMax

---

*Worldwide Interoperability for Microwave Access (WiMAX) is an IEEE standard 802.16 (for fixed wireless) and 802.16e (for mobile wireless) that aims to provide the “last mile” broadband wireless access alternative to cable modem, telephone DSL service..*



## 6.1.4 (continued)

---

### ☐ *Architecture*

- ◆ *Base Station*
- ◆ *Subscriber Stations*
- ◆ *Portable Unit*

### ☐ *Data-Link Layer*

### ☐ *Physical Layer*

### ☐ *Application*

## 6-2 OTHER WIRELESS NETWORKS

*In this section, we concentrate on other wireless networks. We first discuss cellular telephony, which is ubiquitous. We then talk about satellite networks. Before we discuss the above-mentioned wireless networks, let us discuss one access method that we postponed from Chapter 5: channelization, which is used in cellular and other wireless networks.*



## 6.2.1 *Channelization*

---

*Channelization (or channel partition, as it is sometime called) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols: FDMA, TDMA, and CDMA.*

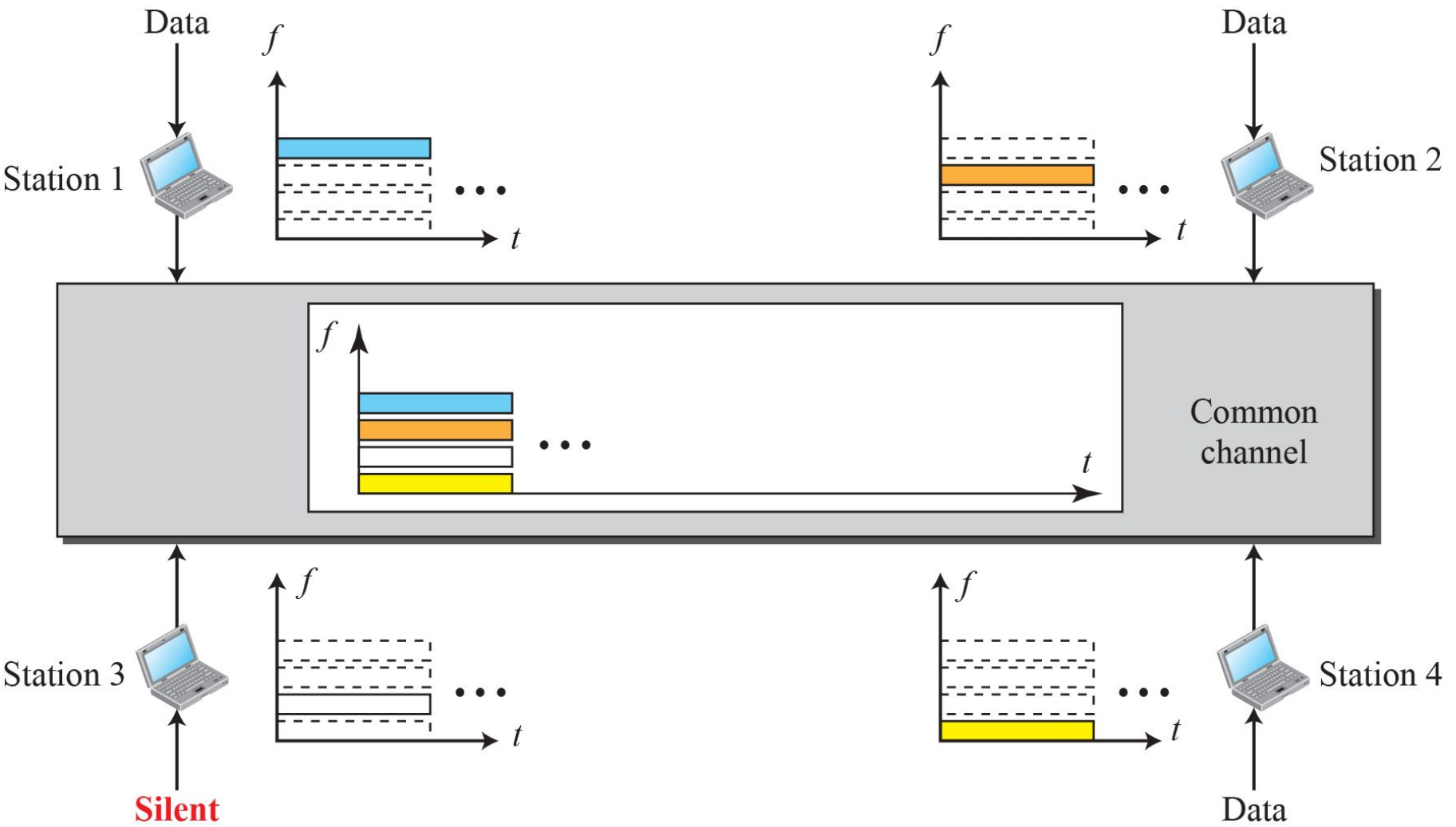


## 6.2.1 (continued)

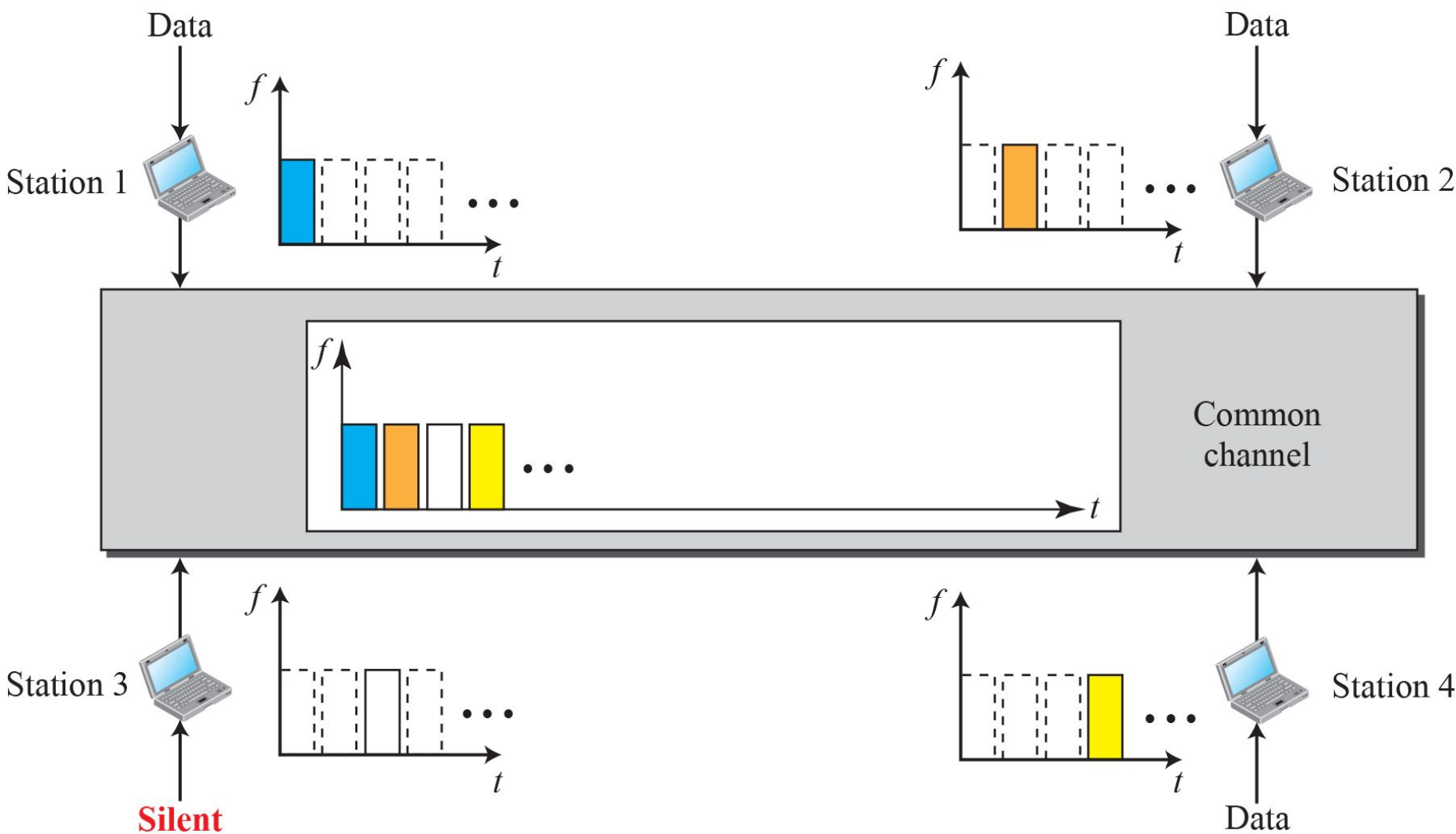
---

- ❑ *Frequency-Division Multiple Access (FDMA)*
- ❑ *Time-Division Multiple Access (TDMA)*
- ❑ *Code-Division Multiple Access (CDMA))*
  - ◆ *Analogy*
  - ◆ *Idea*
  - ◆ *Chips*
  - ◆ *Data Representation*
  - ◆ *Encoding and Decoding*
  - ◆ *Signal Level*
  - ◆ *Sequence Generation*

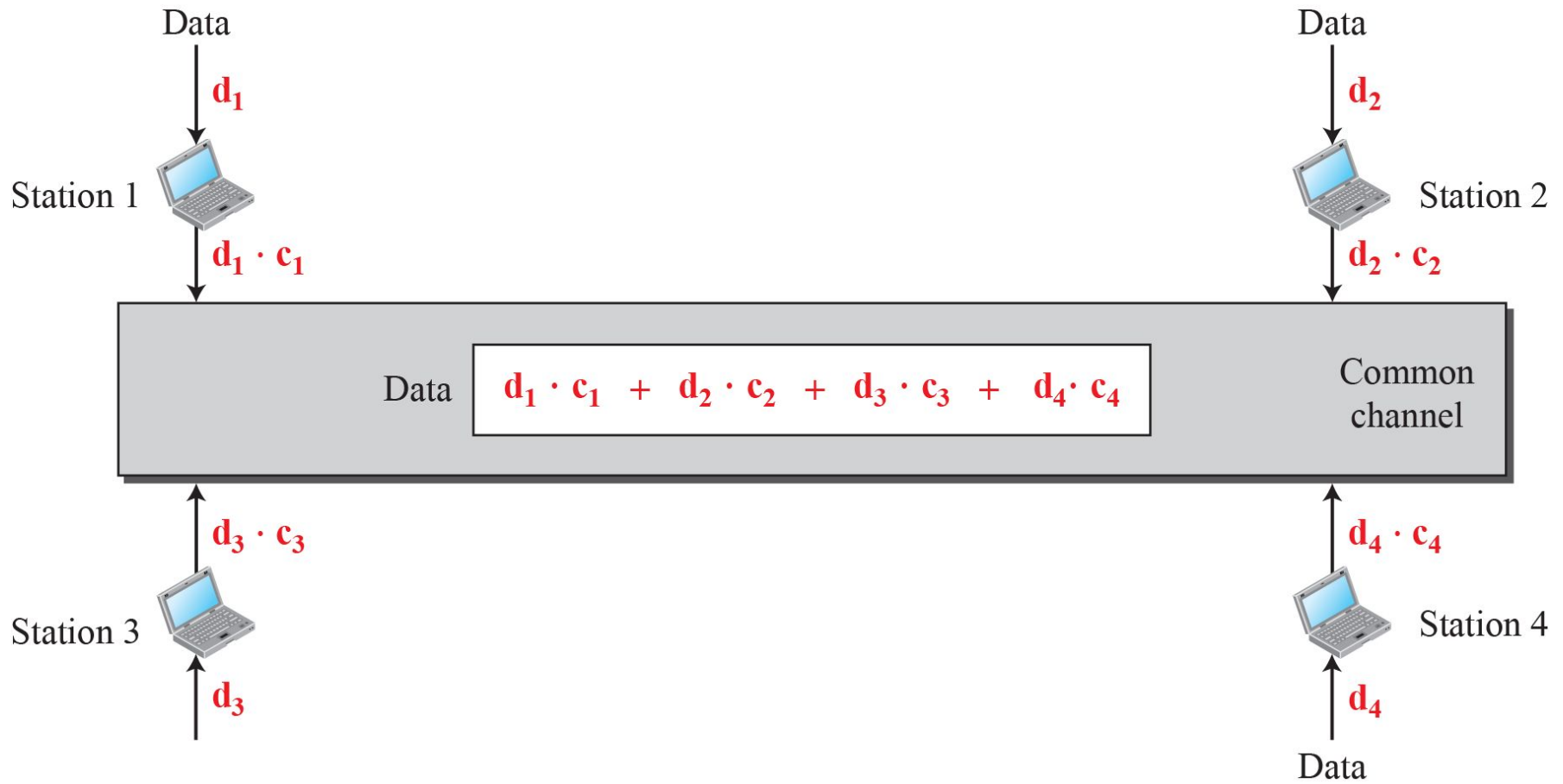
**Figure 6.26:** *Frequency-division multiple access (FDMA)*



**Figure 6.27:** *Time-division multiple access (TDMA)*

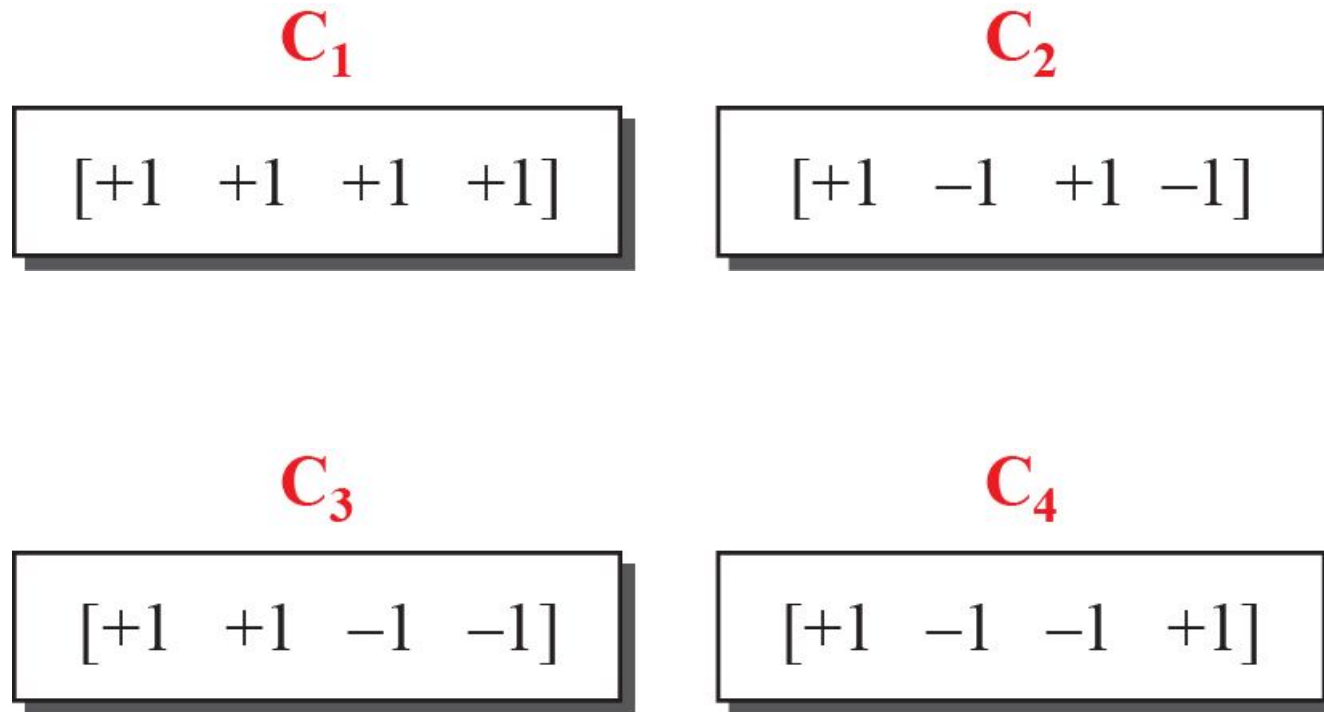


**Figure 6.28:** *Simple idea of communication with code*

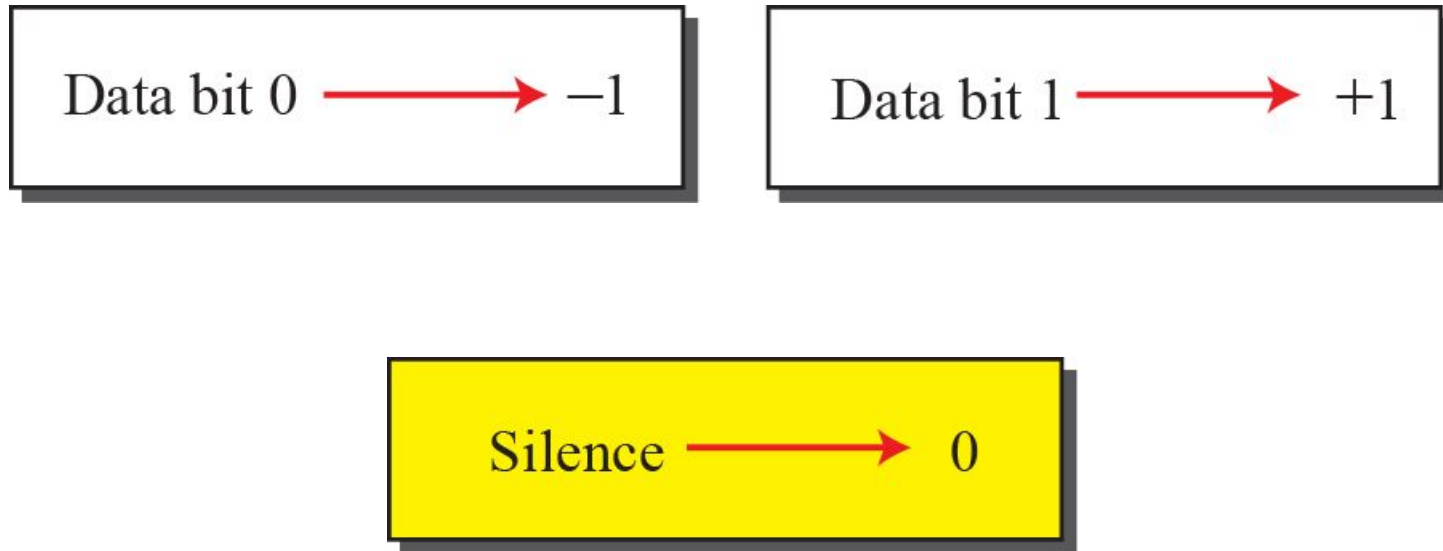




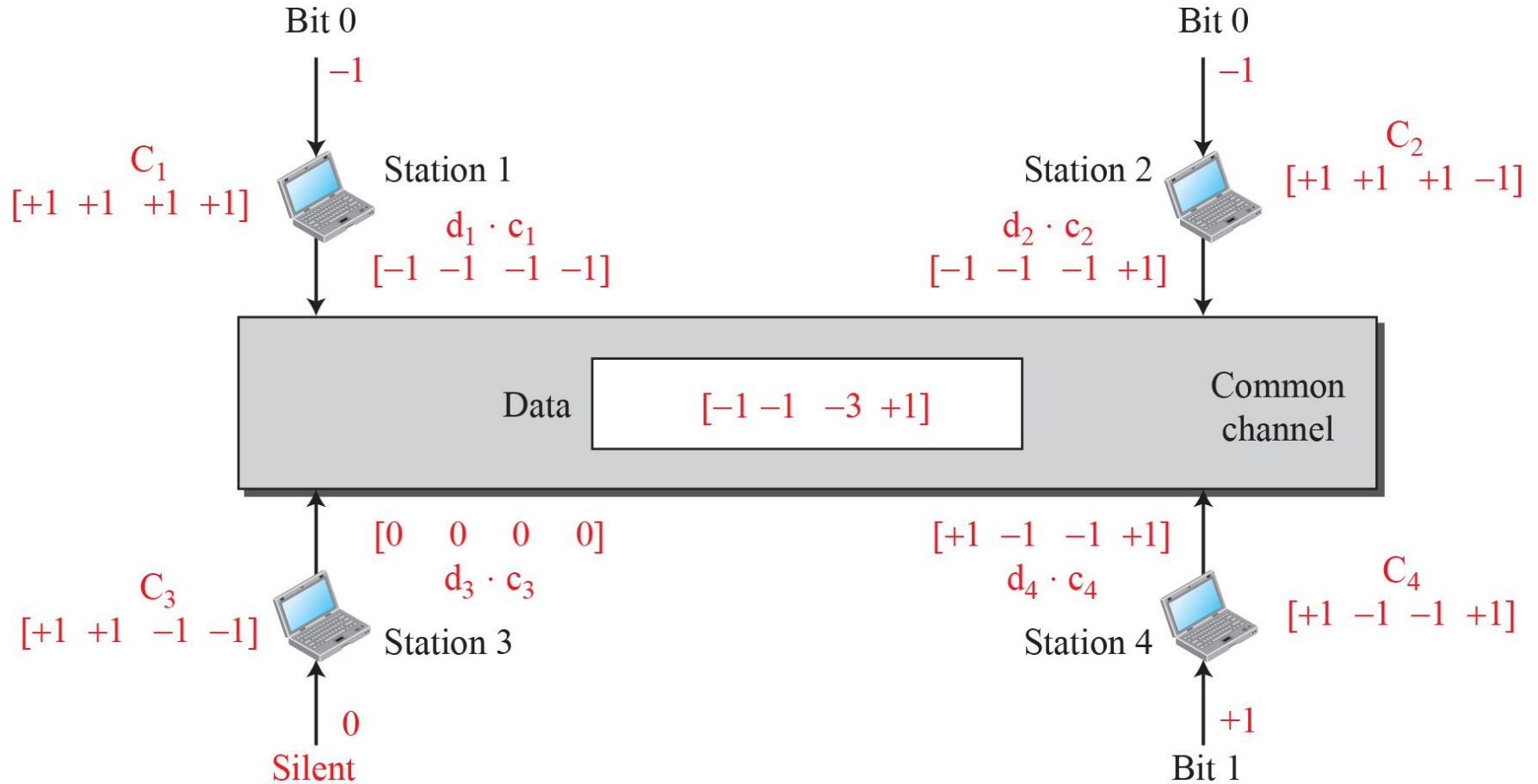
**Figure 6.29:** *Chip sequences*



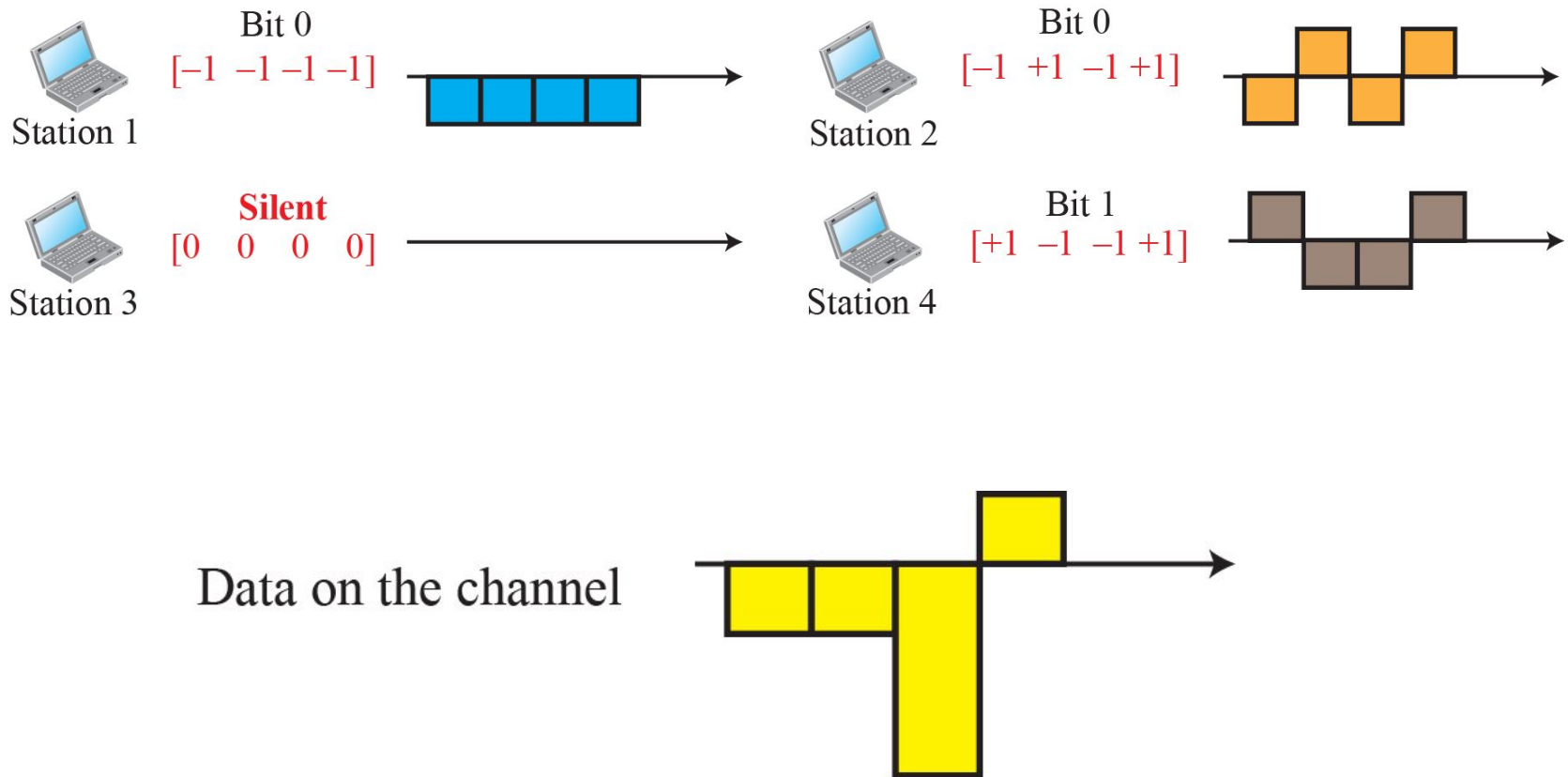
**Figure 6.30:** *Data representation in CDMA*



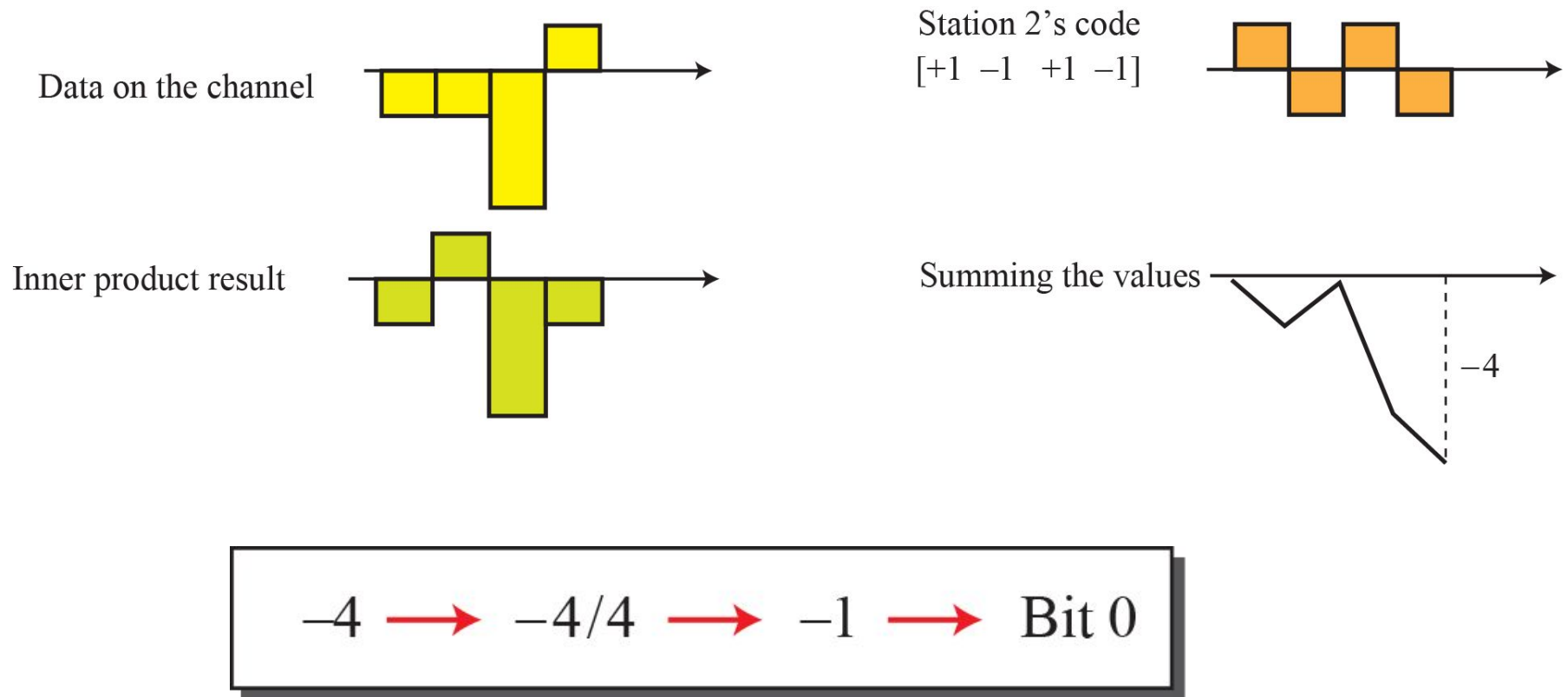
**Figure 6.31: Sharing channel in CDMA**



**Figure 6.32:** *Digital signal created by four stations in CDMA*



**Figure 6.33:** *Decoding of the composite signal for one in CDMA*



**Figure 6.34:** *General rules and examples of creating Walsh tables*

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \quad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_2 = \begin{bmatrix} \begin{bmatrix} +1 & +1 \end{bmatrix} \\ \begin{bmatrix} +1 & -1 \end{bmatrix} \end{bmatrix} \quad W_4 = \begin{bmatrix} \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} & \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \\ \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} & \begin{bmatrix} -1 & -1 \\ -1 & +1 \end{bmatrix} \end{bmatrix}$$

b. Generation of  $W_1$ ,  $W_2$ , and  $W_4$

## Example 6.1

Find the chips for a network with

- a. Two stations
- b. Four stations

### Solution

We can use the rows of  $W_2$  and  $W_4$  in Figure 6.34:

- a. For a two-station network, we have  $[+1 \ +1]$  and  $[+1 \ -1]$ .
- b. For a four-station network we have  $[+1 \ +1 \ +1 \ +1]$ ,  $[+1 \ -1 \ +1 \ -1]$ ,  $[+1 \ +1 \ -1 \ -1]$ , and  $[+1 \ -1 \ -1 \ +1]$ .

## Example 6.2

What is the number of sequences if we have 90 stations in our network?

### **Solution**

The number of sequences needs to be  $2^m$ . We need to choose  $m = 7$  and  $N = 2^7$  or 128. We can then use 90 of the sequences as the chips.



## Example 6.3

Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.

### Solution

Let us prove this for the first station, using our previous four-station example. We can say that the data on the channel  $D = (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4)$ . The receiver, which wants to get the data sent by station 1, multiplies these data by  $c_1$ .

$$\begin{aligned}[D \cdot c_1] / 4 &= [(d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1] / 4 \\ &= [d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1] / 4 \\ &= [d_1 \times 4 + d_2 \times 0 + d_3 \times 0 + d_4 \times 0] / 4 = [d_1 \times 4] / 4 = d_1\end{aligned}$$



## 6.2.2 Cellular Telephony

---

*Cellular telephony is designed to provide communications between two moving units, called mobile stations (MSs), or between one mobile unit and one stationary unit, often called a land unit. A service provider must be able to locate and track a caller, assign a channel to the call, and transfer the channel from base station to base station as the caller moves out of range.*



## 6.2.2 (continued)

---

- *Frequency-Reuse Principle*

- *Transmitting*

- *Receiving*

  - ◆ *Handoff*

- *Roaming*

- *First Generation (1G)*

  - ◆ *AMPS*



## 6.2.2 (continued)

---

### *Second Generation (2G)*

- ◆ *D-AMPS*
- ◆ *GSM*
- ◆ *IS-95*

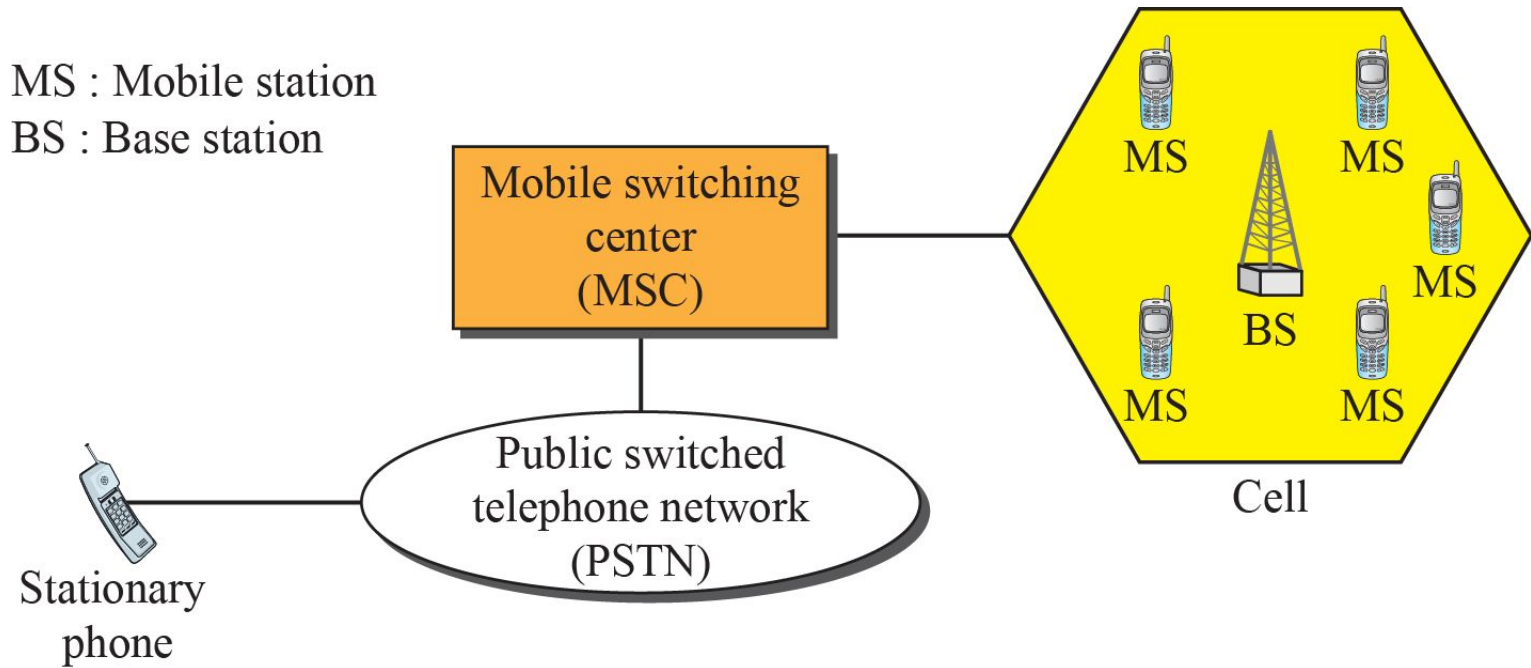
### *Third Generation (3G)*

- ◆ *IMT-2000 Radio Interface*

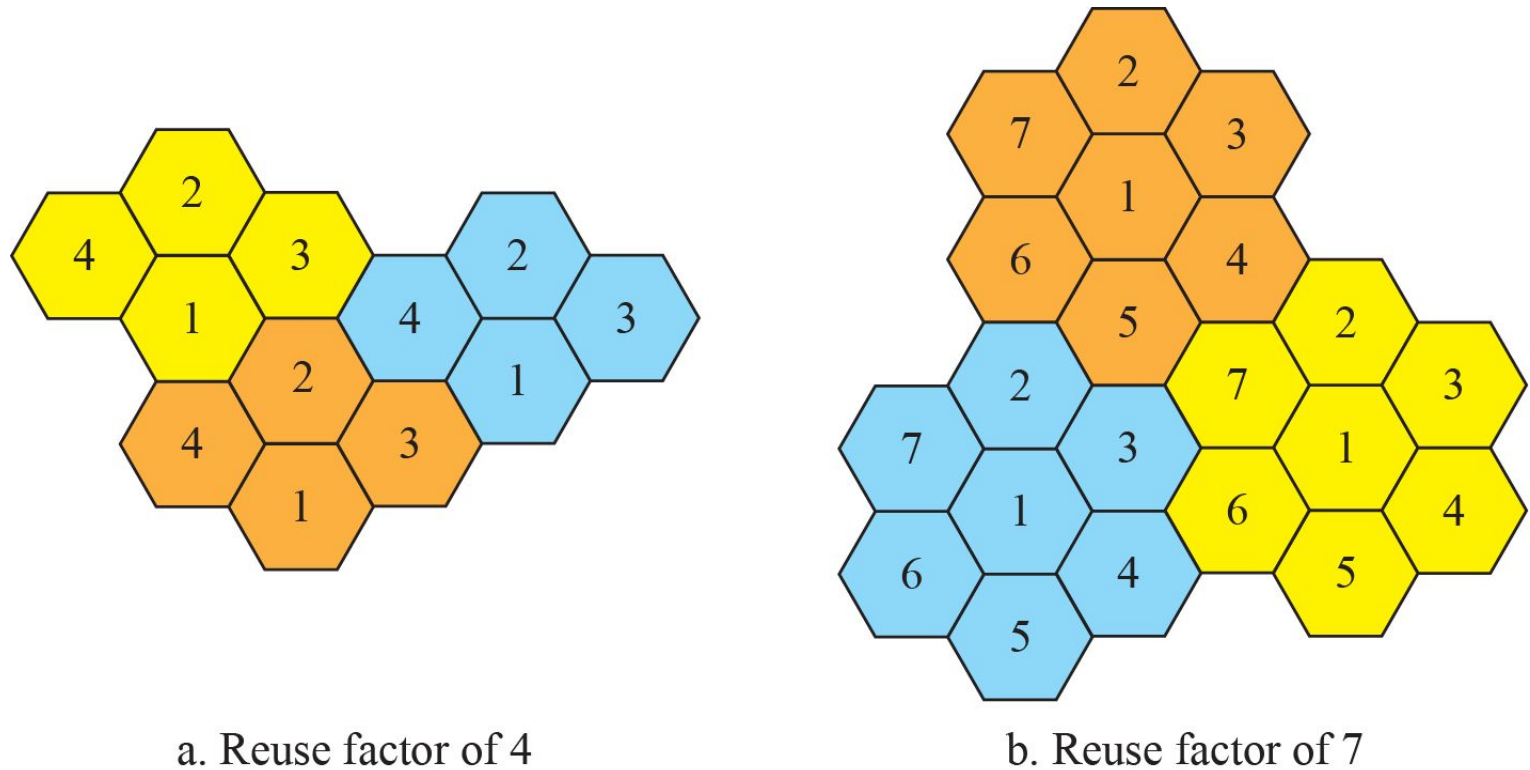
### *Fourth Generation (4G)*

- ◆ *Access Scheme*
- ◆ *Modulation*
- ◆ *Radio System*
- ◆ *Antenna*
- ◆ *Applications*

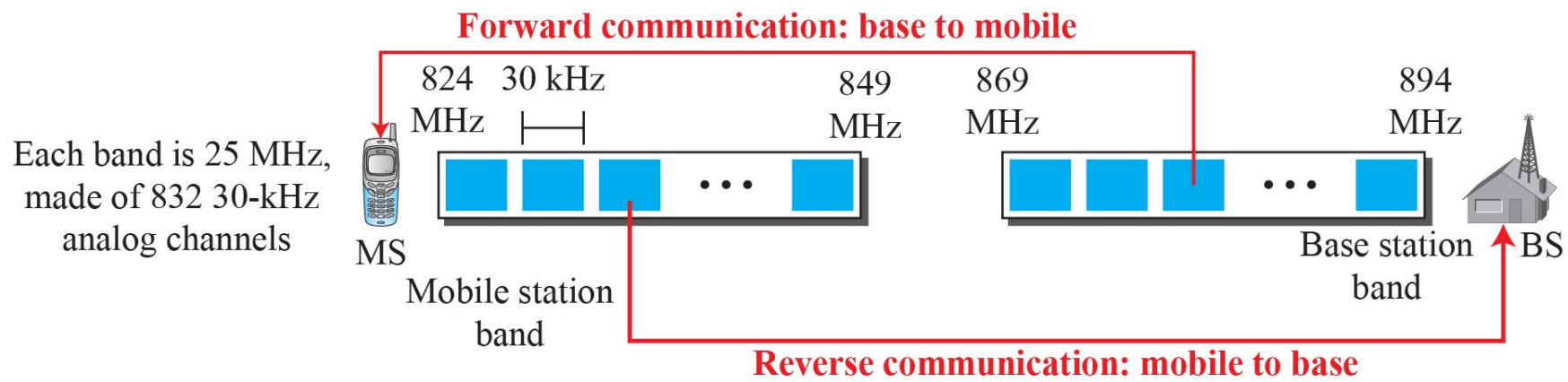
**Figure 6.35: Cellular system**



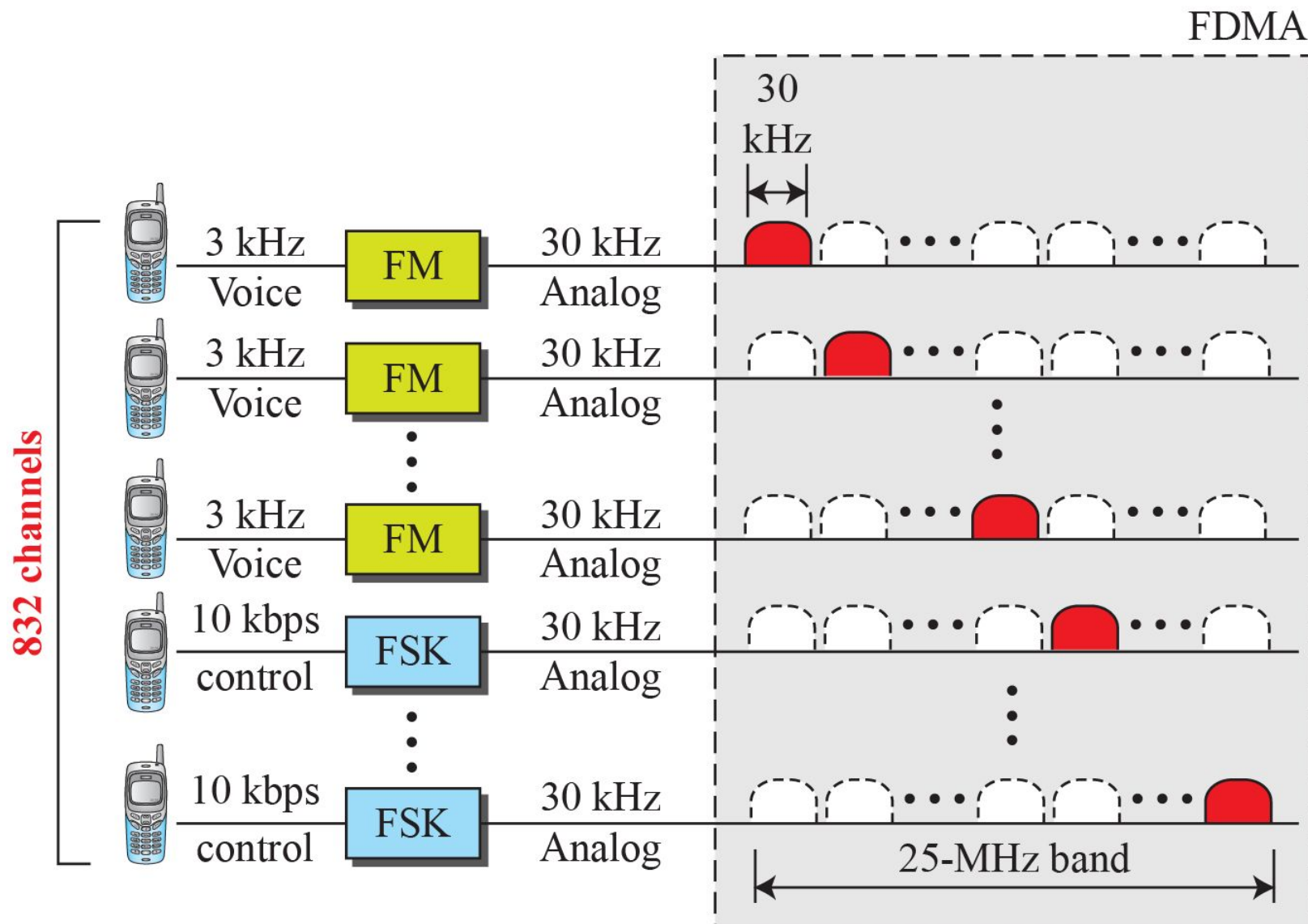
**Figure 6.36:** *Frequency reuse patterns*



**Figure 6.37:** Cellular bands for AMPS

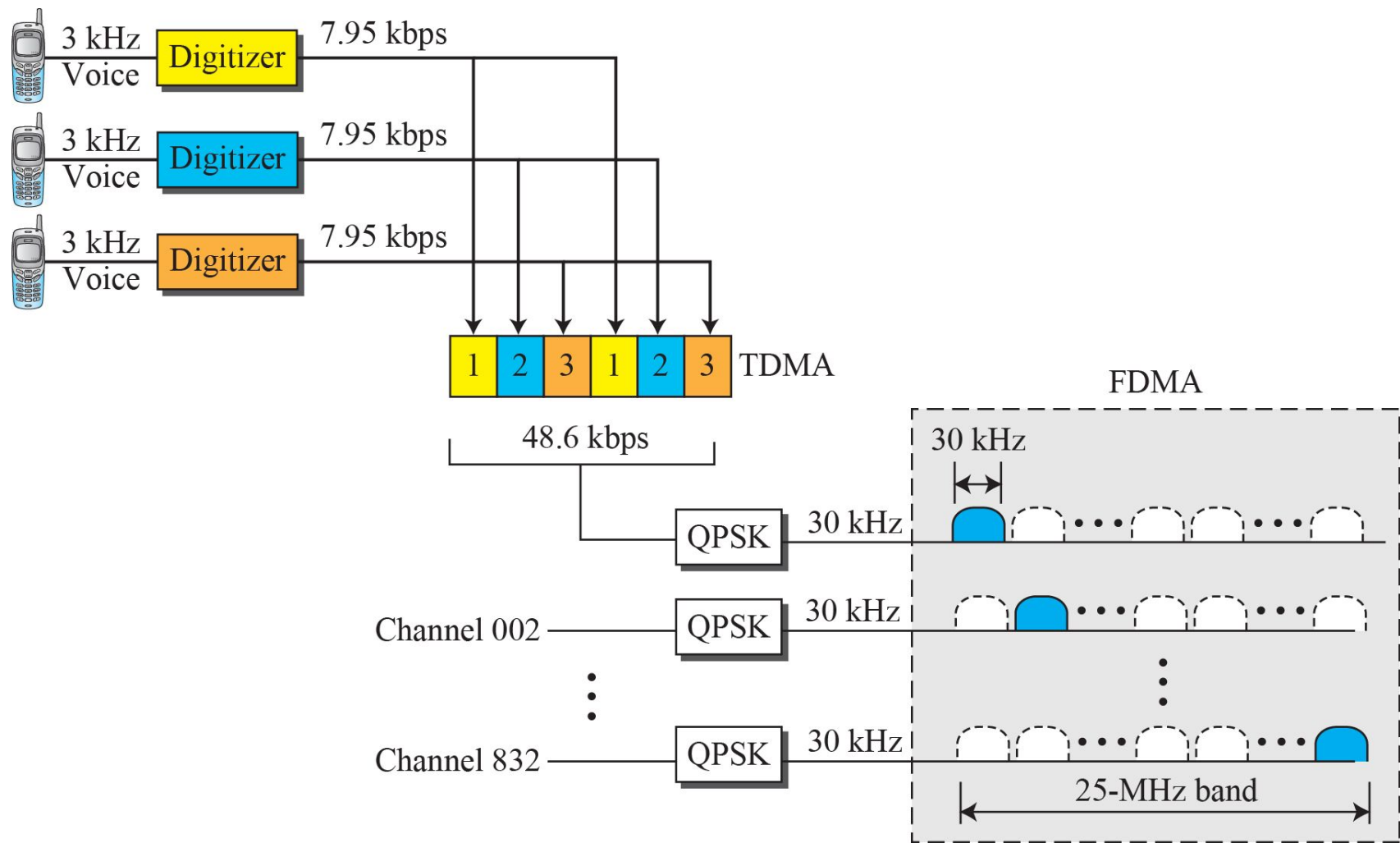


**Figure 6.38:** *AMPS reverse communication band*

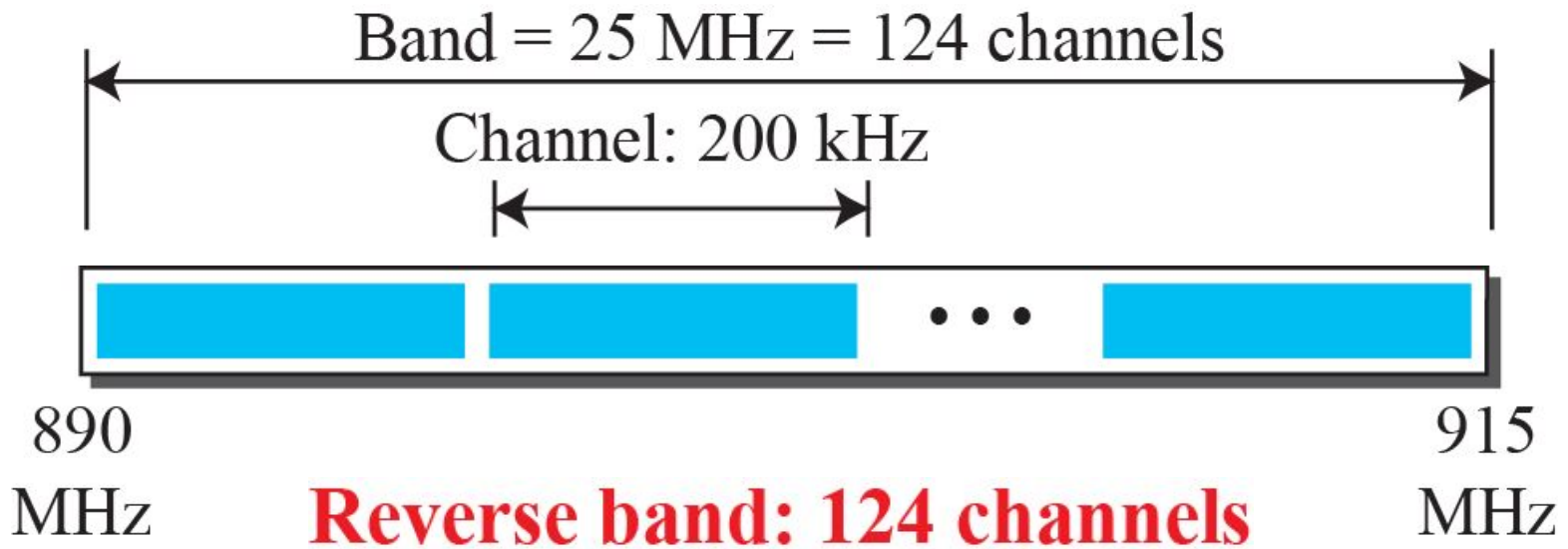




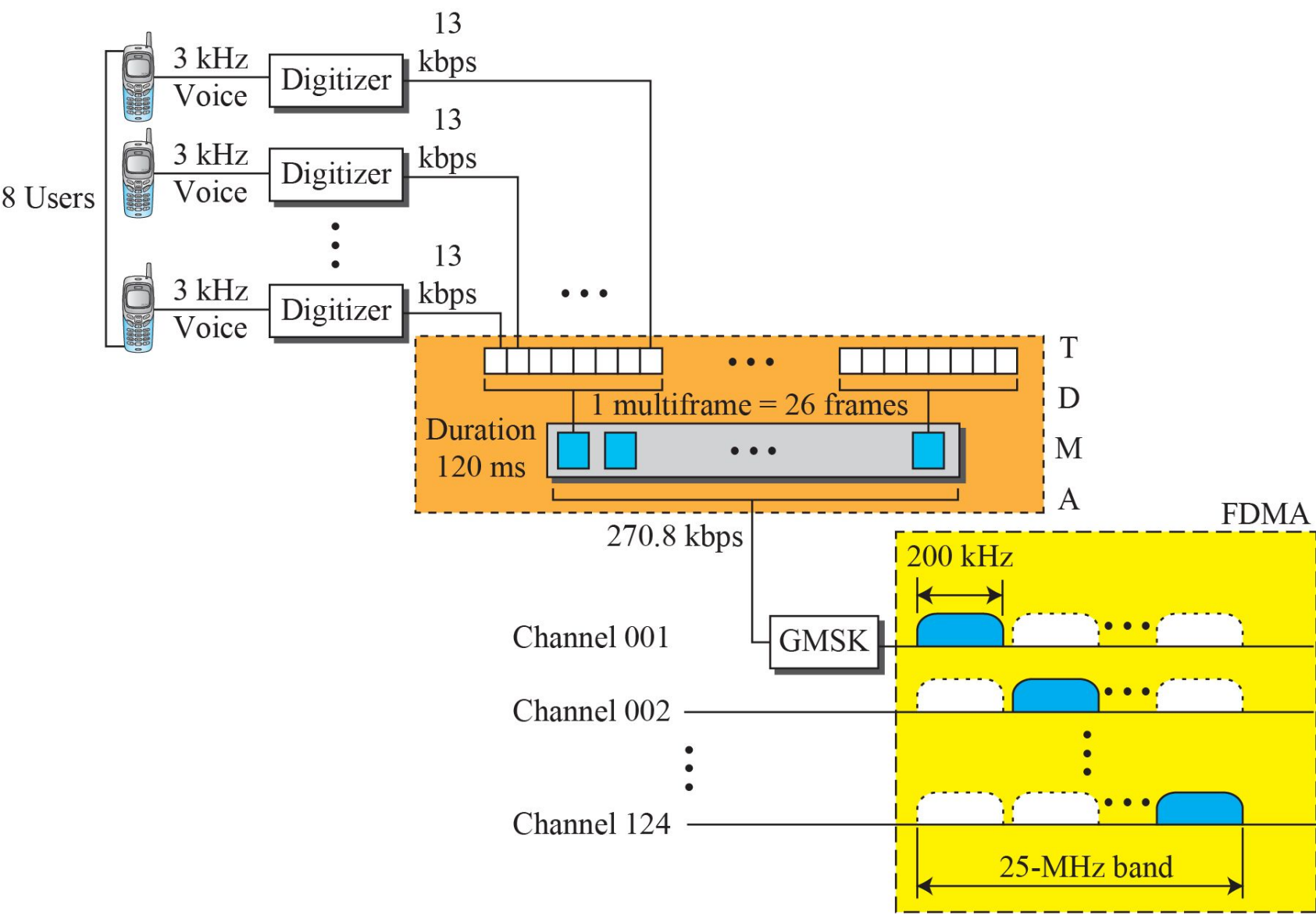
**Figure 6.39: D-AMPS**



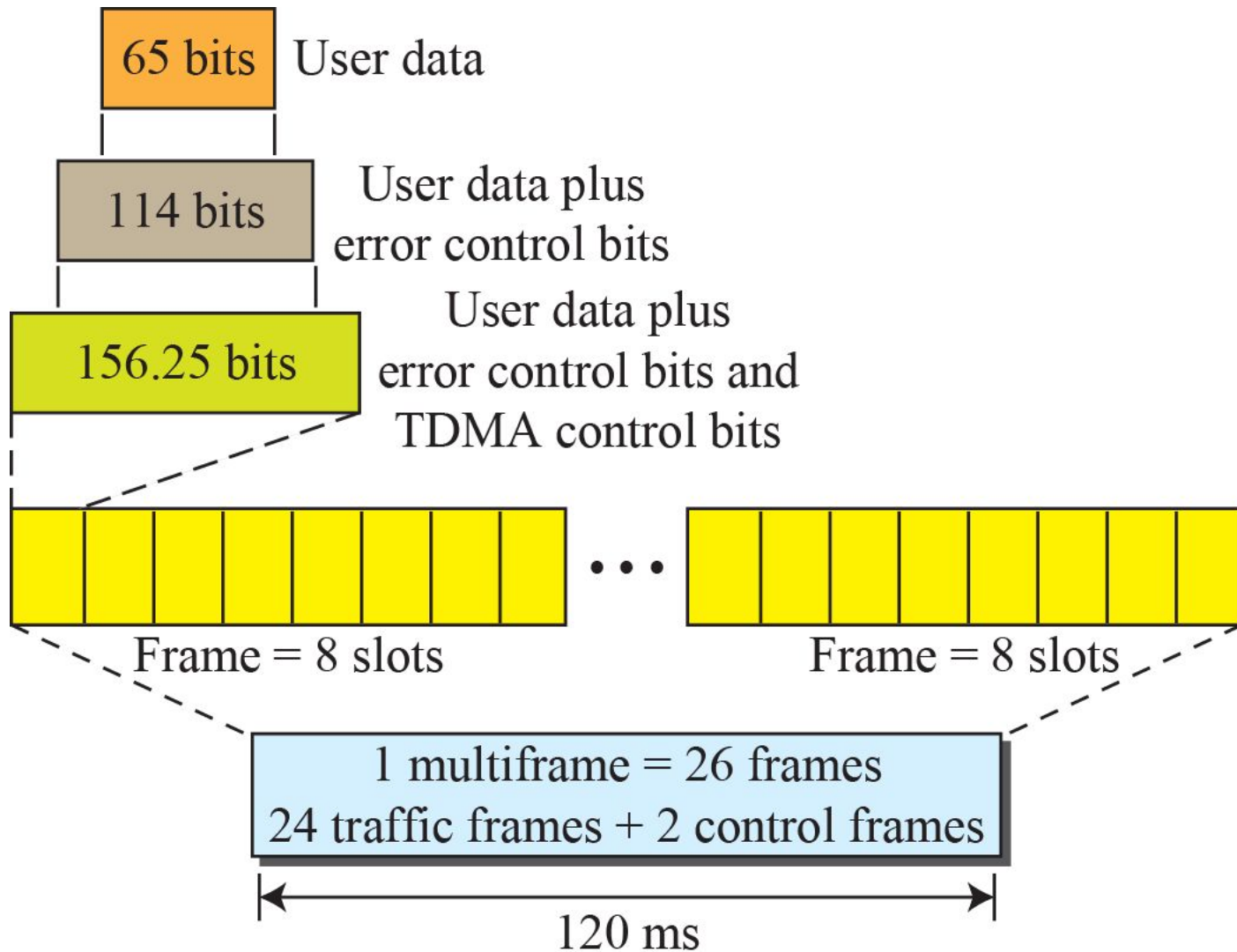
**Figure 6.40: GSM bands**



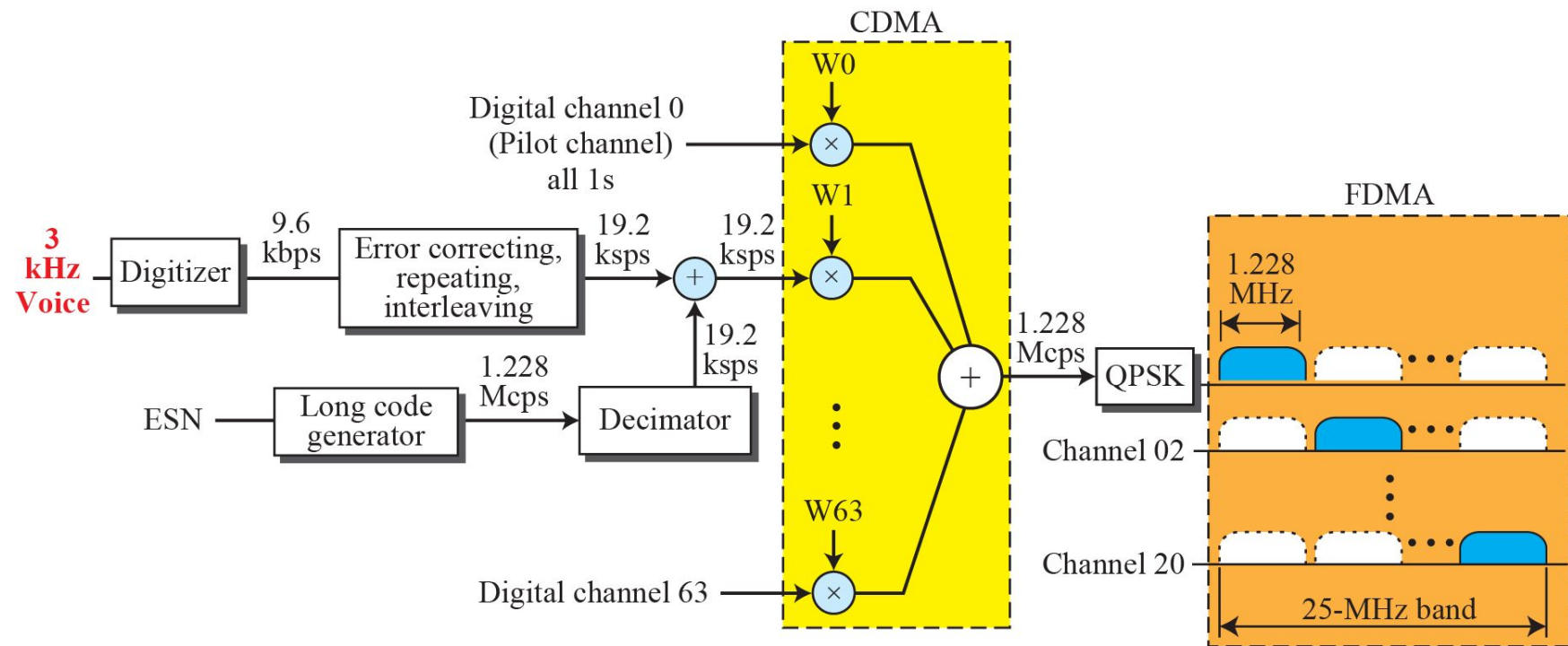
**Figure 6.41: GSM**



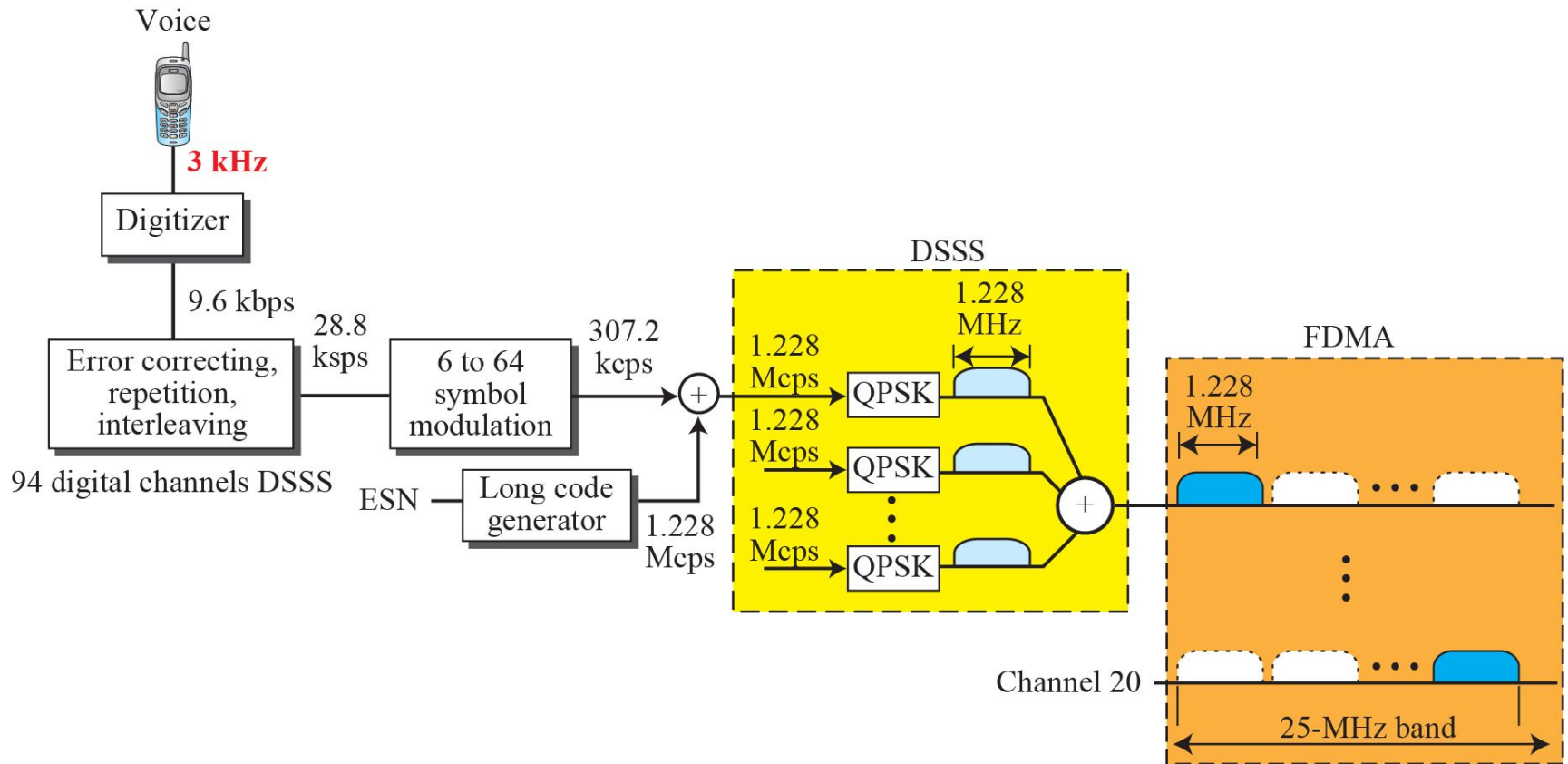
**Figure 6.42:** *Multiframe components*



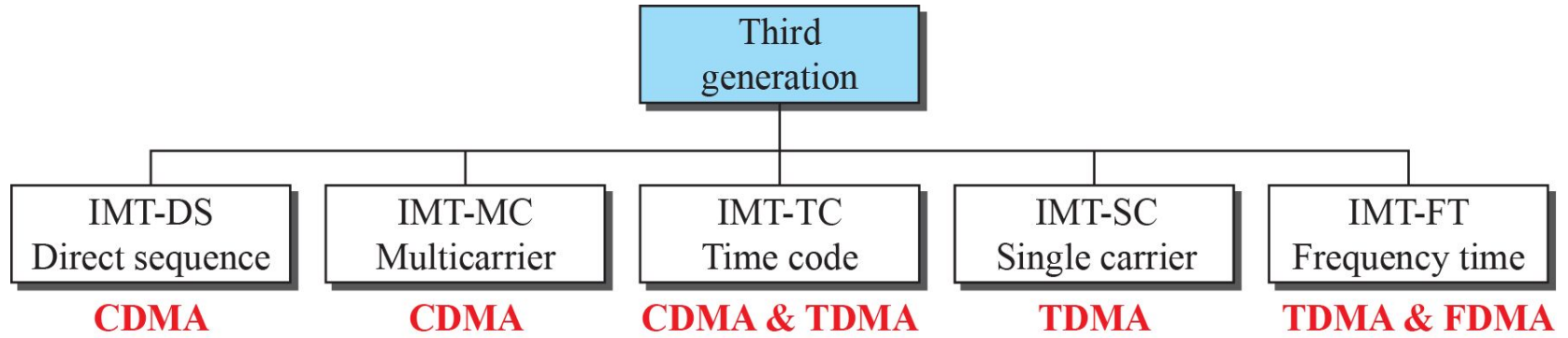
**Figure 6.43:** *IS-95 forward transmission*



**Figure 6.44:** *S-95 reverse transmission*



**Figure 6.45:** *IMT-2000 radio interfaces*





### 6.2.3 *Satellite Networks*

---

*A satellite network is a combination of nodes, some of which are satellites, that provides communication from one point on the Earth to another. A node in the network can be a satellite, an Earth station, or an end-user terminal or telephone.*





## 6.2.3 (continued)

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□ *Orbits*

□ *Footprint*

□ *Three Categories of Satellites*

◆ *Frequency Bands for Satellite Communication*

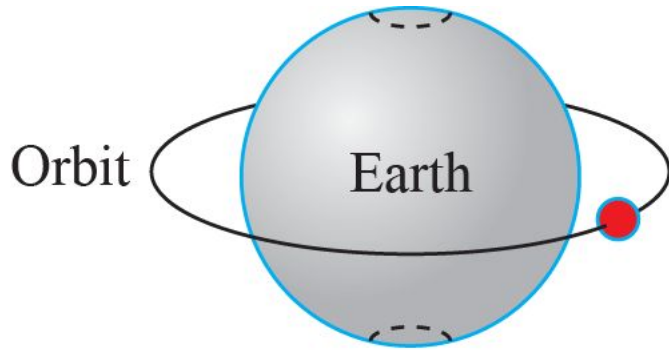
□ *GEO Satellites*

□ *MEO Satellites*

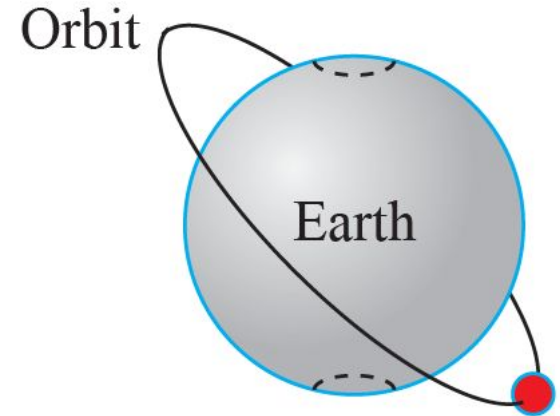
◆ *Global Positioning System (GPS)*

□ *LEO Satellites*

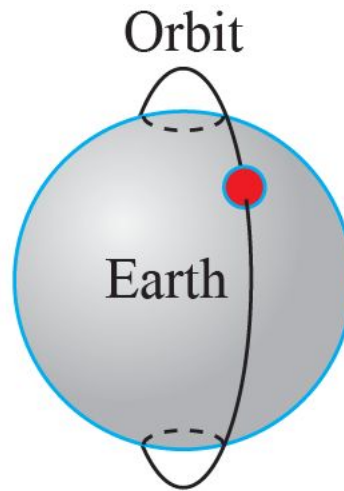
**Figure 6.46:** *Satellite orbits*



a. Equatorial-orbit satellite



b. Inclined-orbit satellite



c. Polar-orbit satellite

## Example 6.4

What is the period of the moon, according to Kepler's law?

$$\text{Period} = C \times \text{distance}^{1.5}$$

Here  $C$  is a constant approximately equal to  $1/100$ . The period is in seconds and the distance in kilometers.

### Solution

The moon is located approximately 384,000 km above the Earth. The radius of the Earth is 6378 km. Applying the formula, we get the following.

$$\text{Period} = (1/100) \times (384,000 + 6378)^{1.5} = 2,439,090 \text{ s} = 1 \text{ month}$$

## Example 6.5

According to Kepler's law, what is the period of a satellite that is located at an orbit approximately 35,786 km above the Earth?

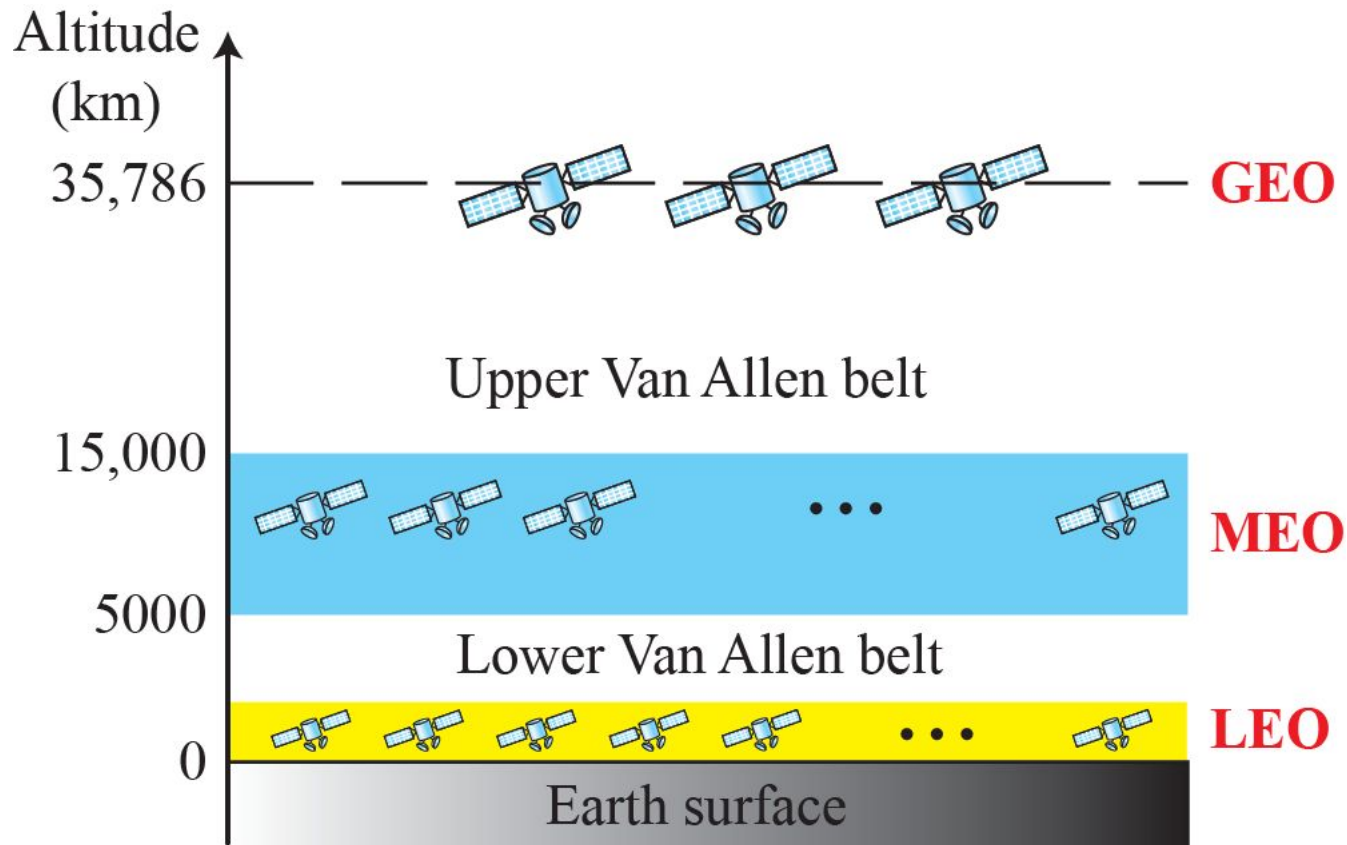
### Solution

Applying the formula, we get the following.

$$\text{Period} = (1/100) \times (35,786 + 6378)^{1.5} = 86,579 \text{ s} = 24 \text{ h}$$

This means that a satellite located at 35,786 km has a period of 24 h, which is the same as the rotation period of the Earth. A satellite like this is said to be *stationary* to the Earth. The orbit, as we will see, is called a *geostationary orbit*.

**Figure 6.47:** *Satellite orbit altitudes*

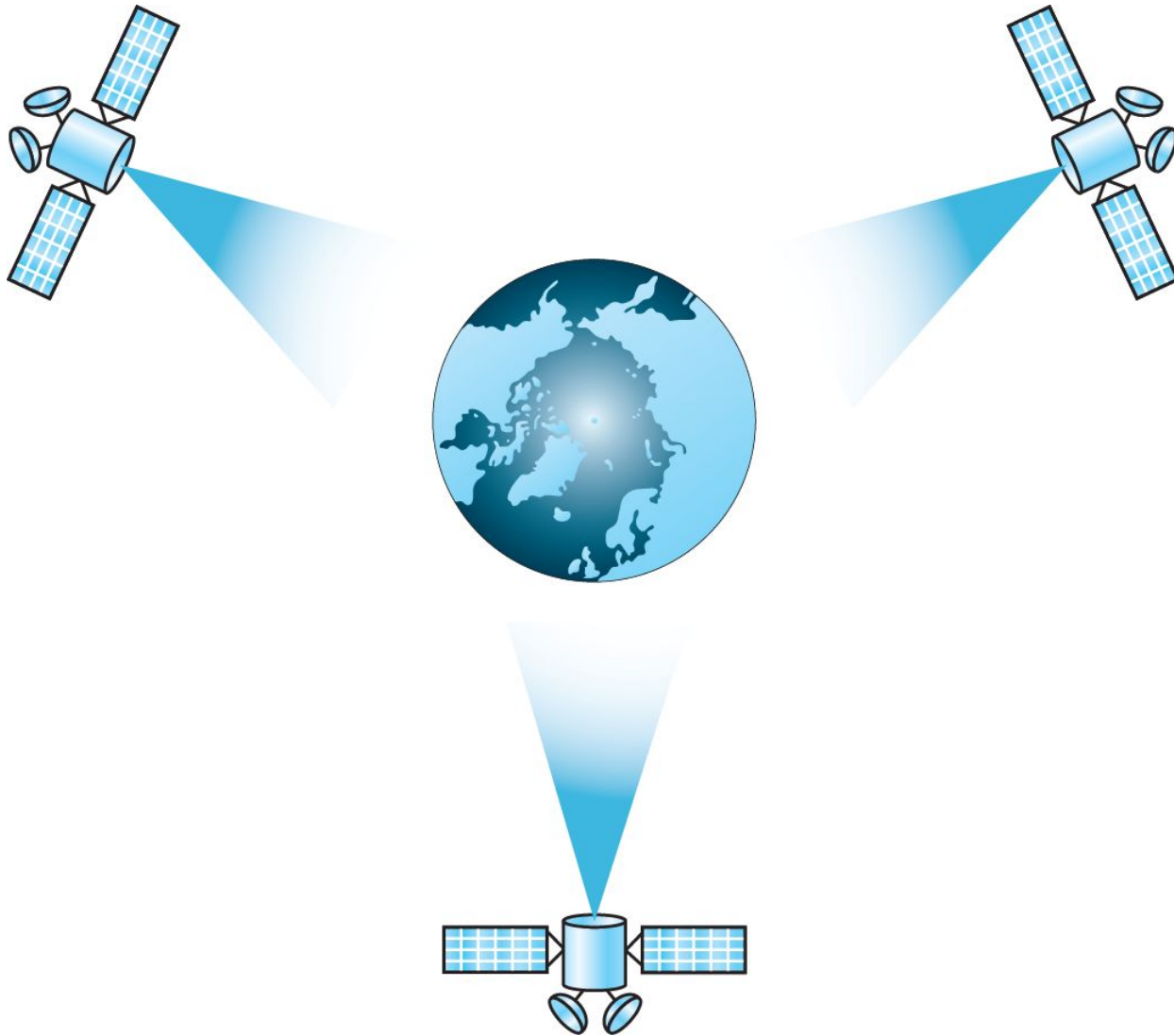




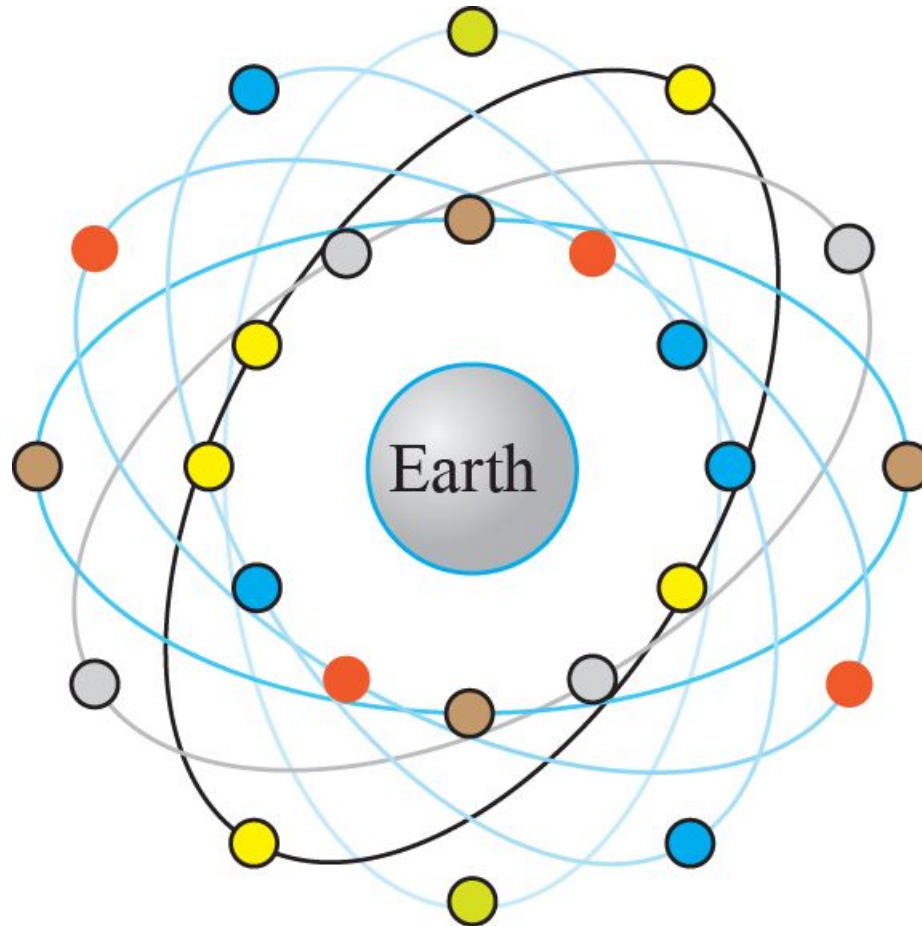
**Table 6.5:** Satellite frequency bands

| <i>Band</i> | <i>Downlink, GHz</i> | <i>Uplink, GHz</i> | <i>Bandwidth, MHz</i> |
|-------------|----------------------|--------------------|-----------------------|
| L           | 1.5                  | 1.6                | 15                    |
| S           | 1.9                  | 2.2                | 70                    |
| C           | 4.0                  | 6.0                | 500                   |
| Ku          | 11.0                 | 14.0               | 500                   |
| Ka          | 20.0                 | 30.0               | 3500                  |

**Figure 6.48:** *Satellites in geostationary orbit*

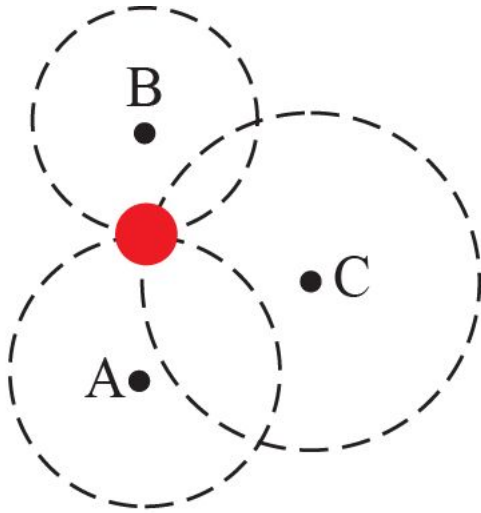


**Figure 6.49:** *Orbits for global positioning system (GPS) satellites*

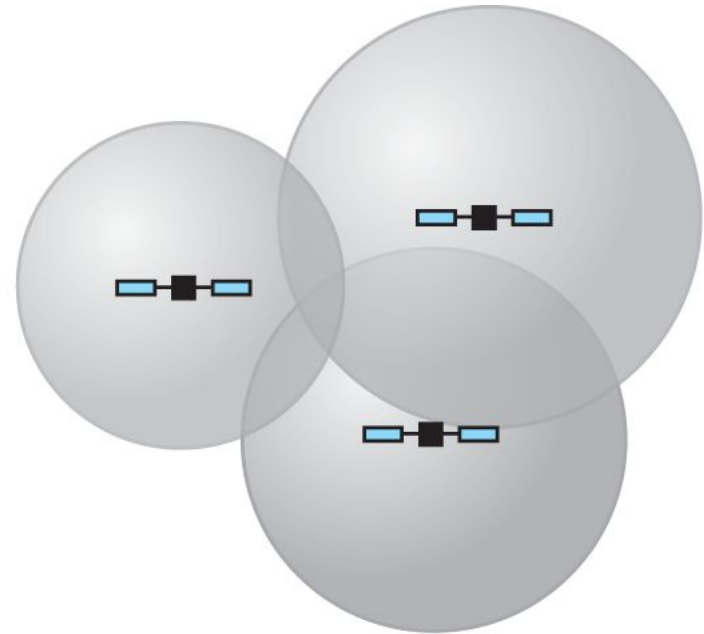




**Figure 6.50:** *Trilateration on a plane*

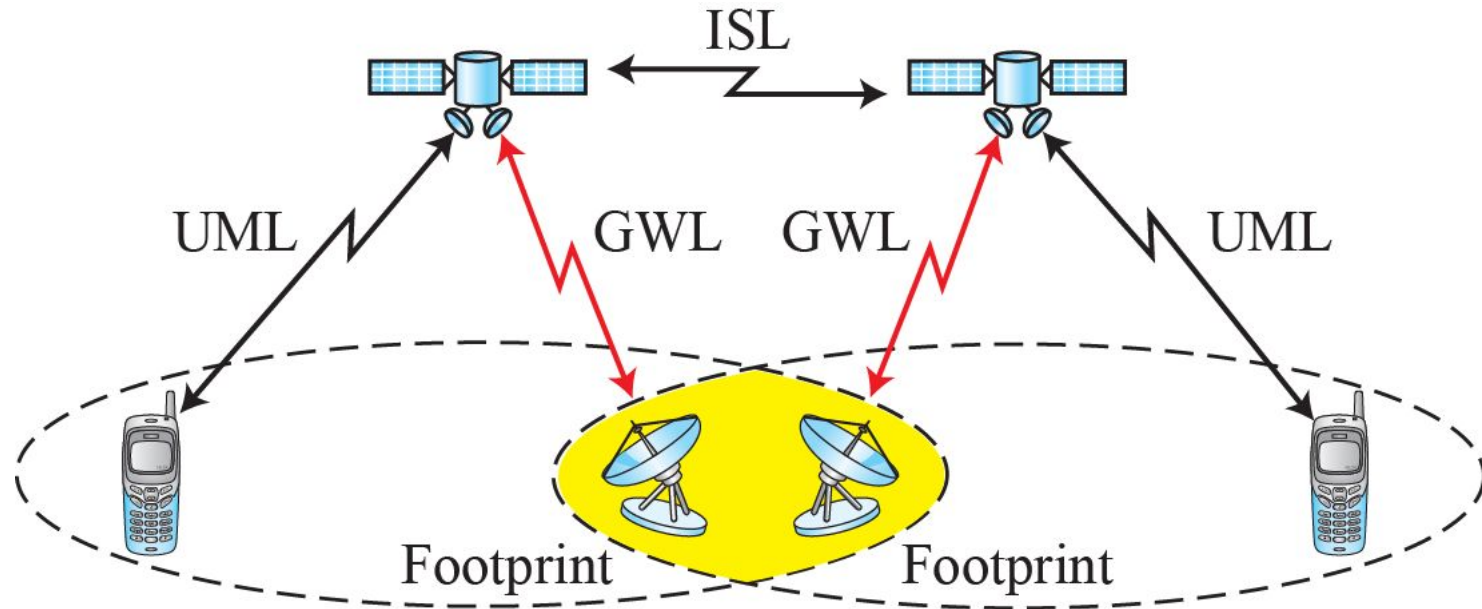


a. Two-dimensional trilateration



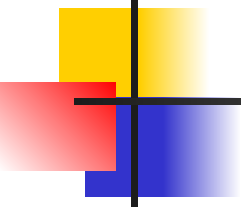
b. Three-dimensional trilateration

**Figure 6.51:** *LEO satellite system*



## 6-3 MOBILE IP

*As mobile and personal computers such as notebooks become increasingly popular, we need to think about mobile IP, the extension of IP protocol that allows mobile computers to be connected to the Internet at any location where the connection is possible. In this section, we discuss this issue.*



## 6.3.1 Addressing

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*The main problem that must be solved in providing mobile communication using the IP protocol is addressing.*

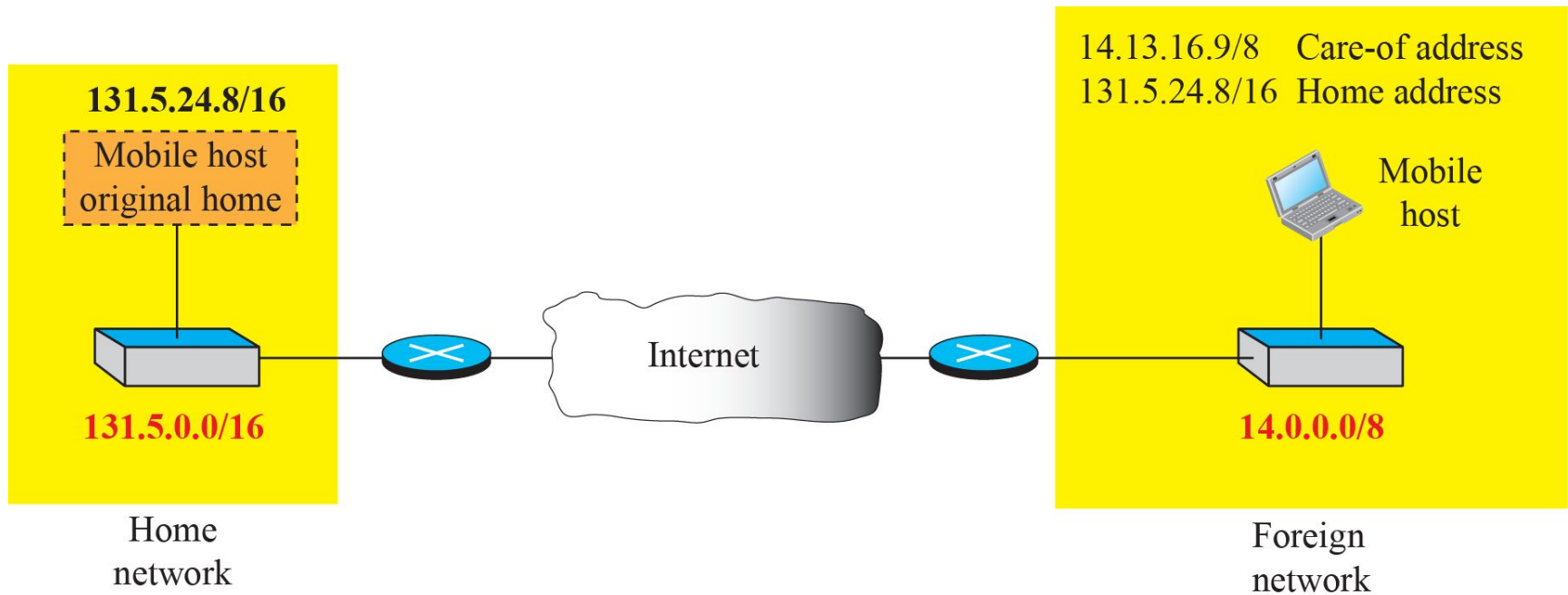
**□ *Stationary Hosts***

**□ *Mobile Hosts***

◆ *Changing the Address*

◆ *Two Addresses*

**Figure 6.52:** *Home address and care-of address*





## 6.3.2 Agents

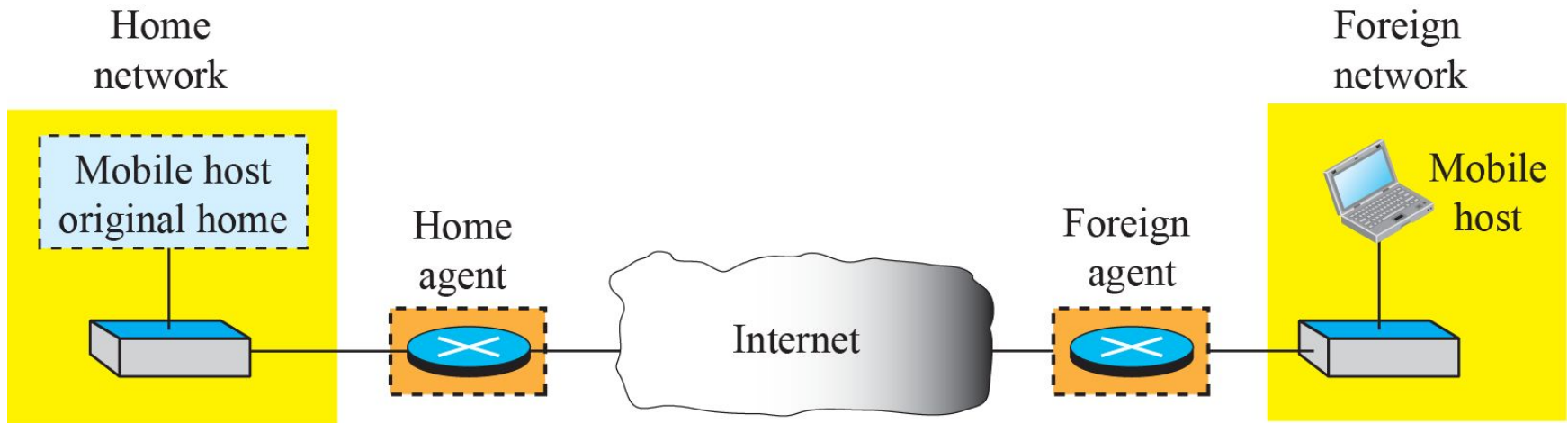
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*To make the change of address transparent to the rest of the Internet requires a home agent and a foreign agent. Figure 6.53 shows the position of a home agent relative to the home network and a foreign agent relative to the foreign network.*

☐ *Home Agent*

☐ *Foreign Agent*

**Figure 6.53:** *Home agent and foreign agent*





### 6.3.3 *Three Phases*

---

*To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer, as shown in Figure 6.54.*

#### *Agent Discovery*

- ◆ *Agent Advertisement*
- ◆ *Agent Solicitation*





## 6.3.3 (continued)

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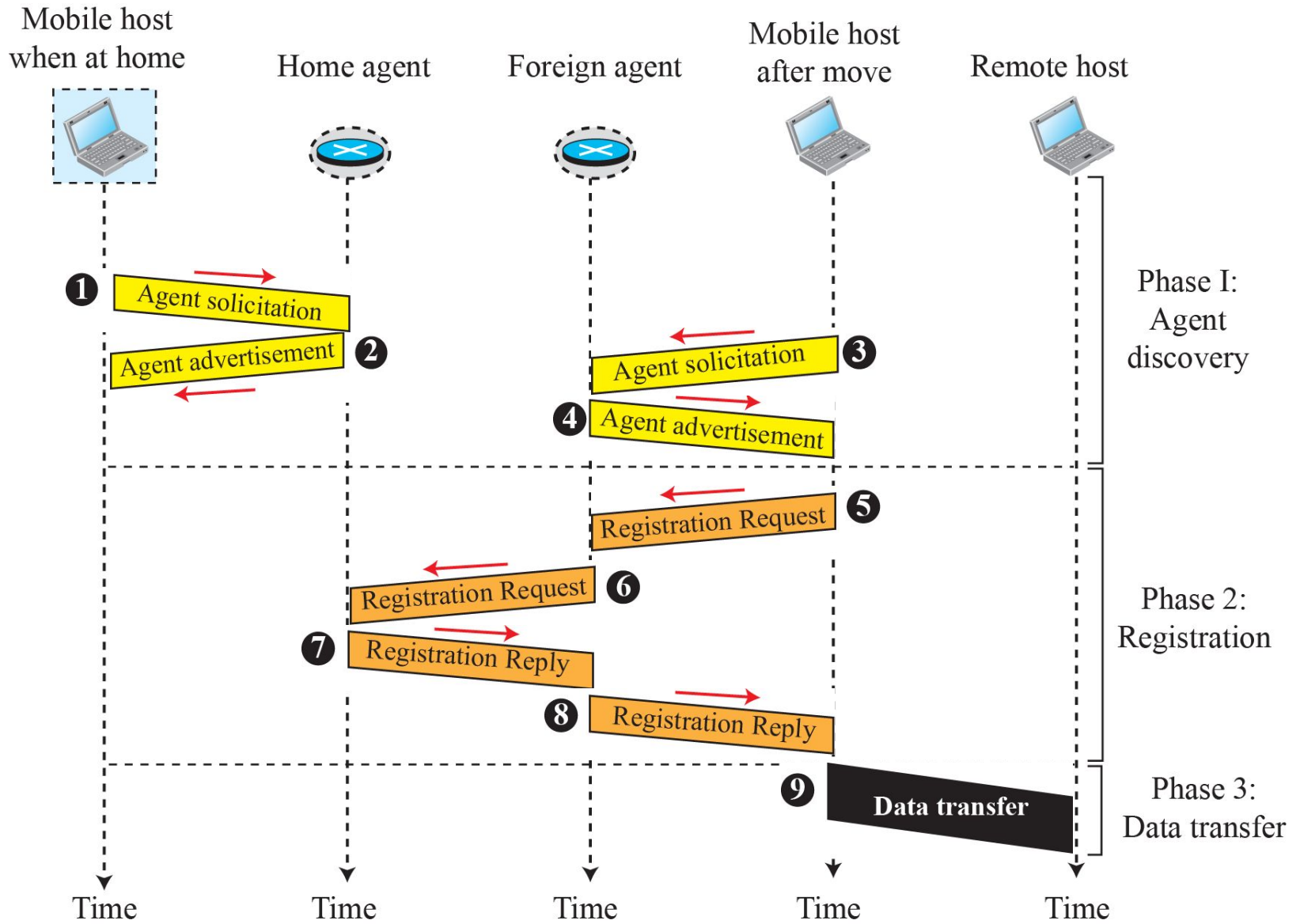
### *Registration*

- ◆ *Request and Reply*
- ◆ *Encapsulation*

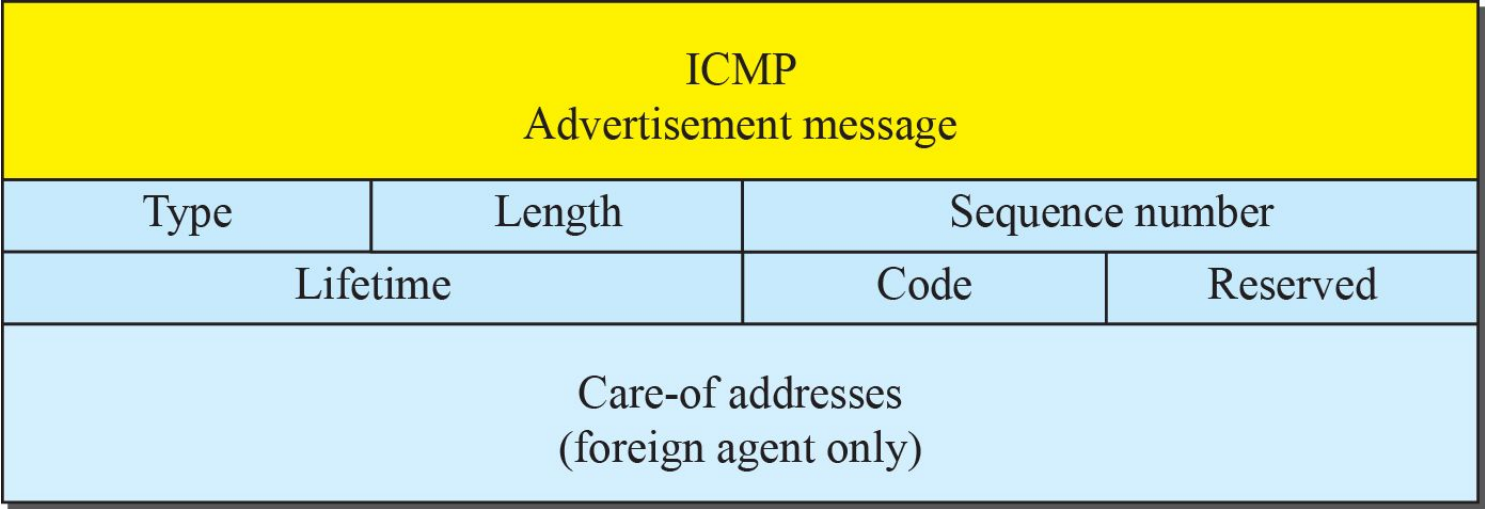
### *Data Transfer*

- ◆ *From Remote Host to Home Agent*
- ◆ *From Home Agent to Foreign Agent*
- ◆ *From Foreign Agent to Mobile Host*
- ◆ *From Mobile Host to Remote Host*
- ◆ *Transparency*

**Figure 6.54: Remote host and mobile host communication**



**Figure 6.55:** *Agent advertisement*





## Table 6.6: Code Bits

| <i>Bit</i> | <i>Meaning</i>                                                 |
|------------|----------------------------------------------------------------|
| 0          | Registration required. No collocated care-of address.          |
| 1          | Agent is busy and does not accept registration at this moment. |
| 2          | Agent acts as a home agent.                                    |
| 3          | Agent acts as a foreign agent.                                 |
| 4          | Agent uses minimal encapsulation.                              |
| 5          | Agent uses generic routing encapsulation (GRE).                |
| 6          | Agent supports header compression.                             |
| 7          | Unused (0).                                                    |

**Figure 6.56:** *Registration request format*

| Type               | Flag | Lifetime |
|--------------------|------|----------|
| Home address       |      |          |
| Home agent address |      |          |
| Care-of address    |      |          |
| Identification     |      |          |
| Extensions ...     |      |          |



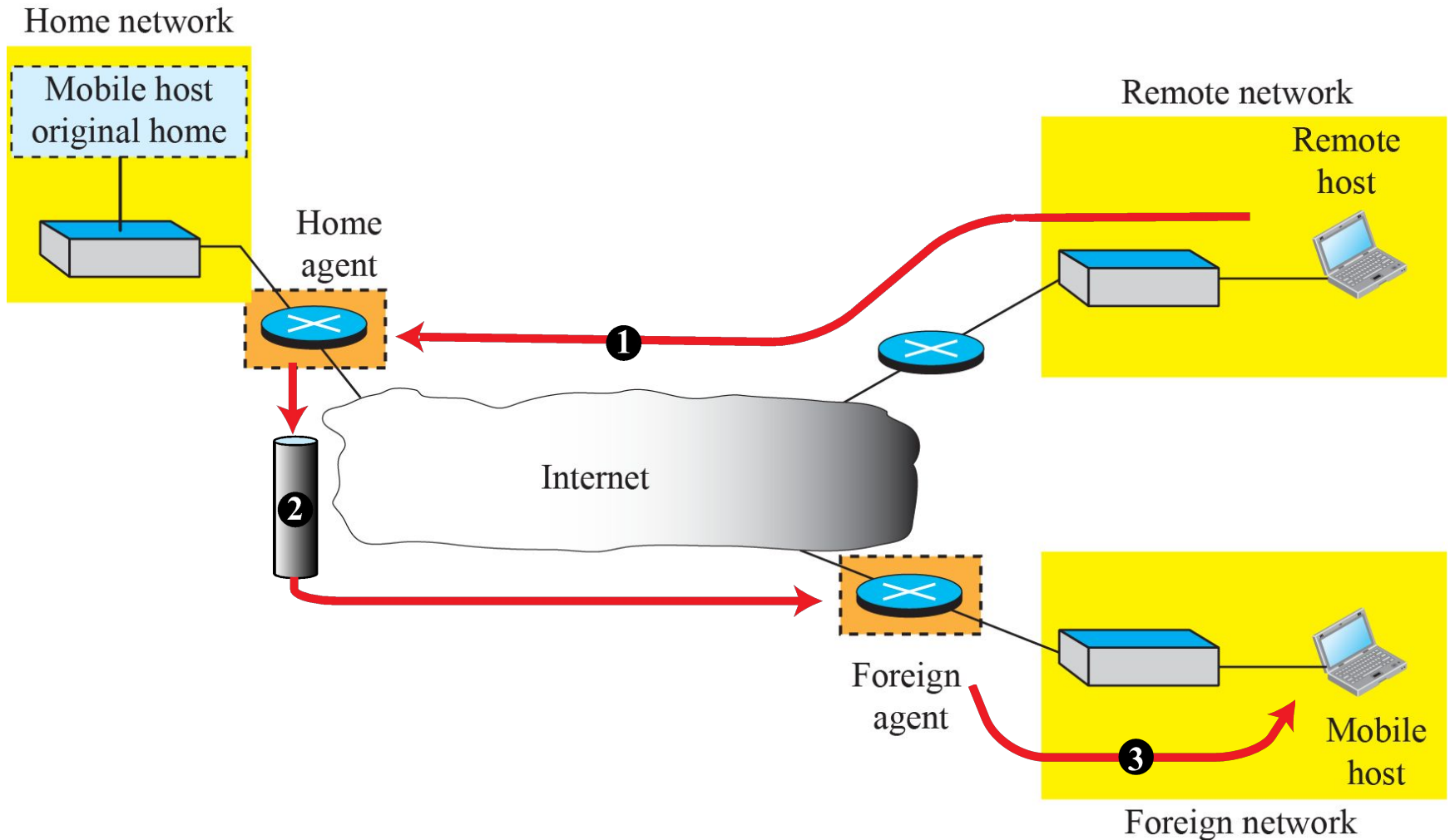
**Table 6.7:** Registration request flag field bits

| <i>Bit</i> | <i>Meaning</i>                                                         |
|------------|------------------------------------------------------------------------|
| 0          | Mobile host requests that home agent retain its prior care-of address. |
| 1          | Mobile host requests that home agent tunnel any broadcast message.     |
| 2          | Mobile host is using collocated care-of address.                       |
| 3          | Mobile host requests that home agent use minimal encapsulation.        |
| 4          | Mobile host requests generic routing encapsulation (GRE).              |
| 5          | Mobile host requests header compression.                               |
| 6–7        | Reserved bits.                                                         |

**Figure 6.57:** *Registration reply format*

| Type               | Code | Lifetime |
|--------------------|------|----------|
| Home address       |      |          |
| Home agent address |      |          |
| Identification     |      |          |
| Extensions ...     |      |          |

**Figure 6.58:** *Data transfer*







### 6.3.4 *Inefficiency in Mobile IP*

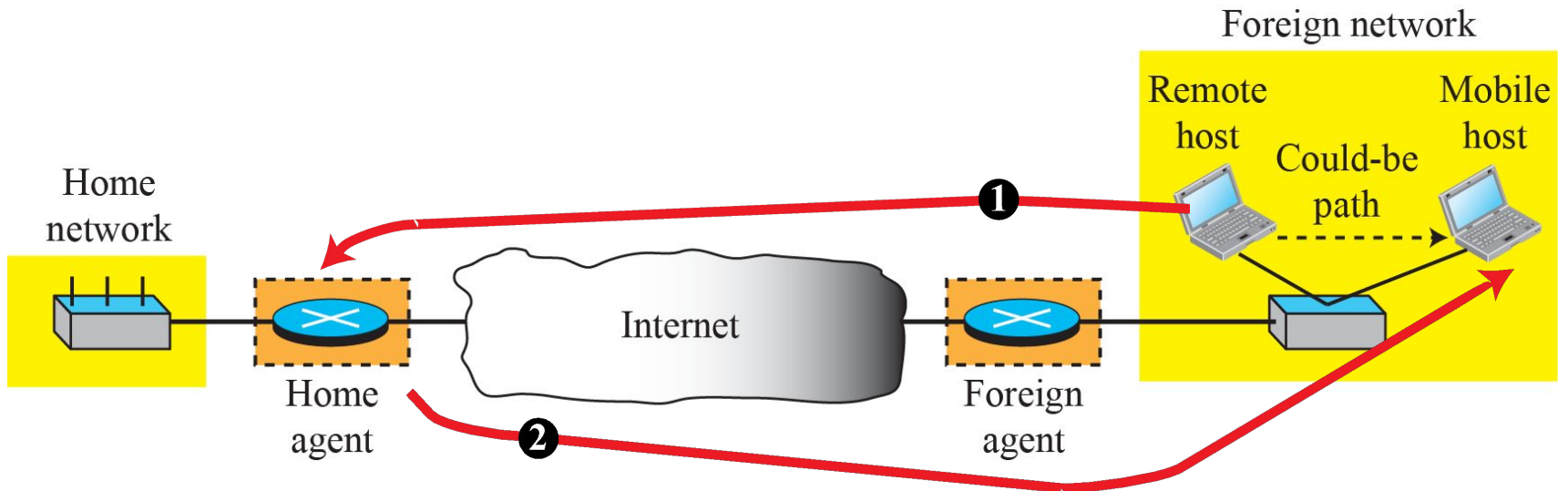
*Communication involving mobile IP can be inefficient. The inefficiency can be severe or moderate. The severe case is called double crossing or 2X. The moderate case is called triangle routing or dog-leg routing.*

**□ *Double Crossing***

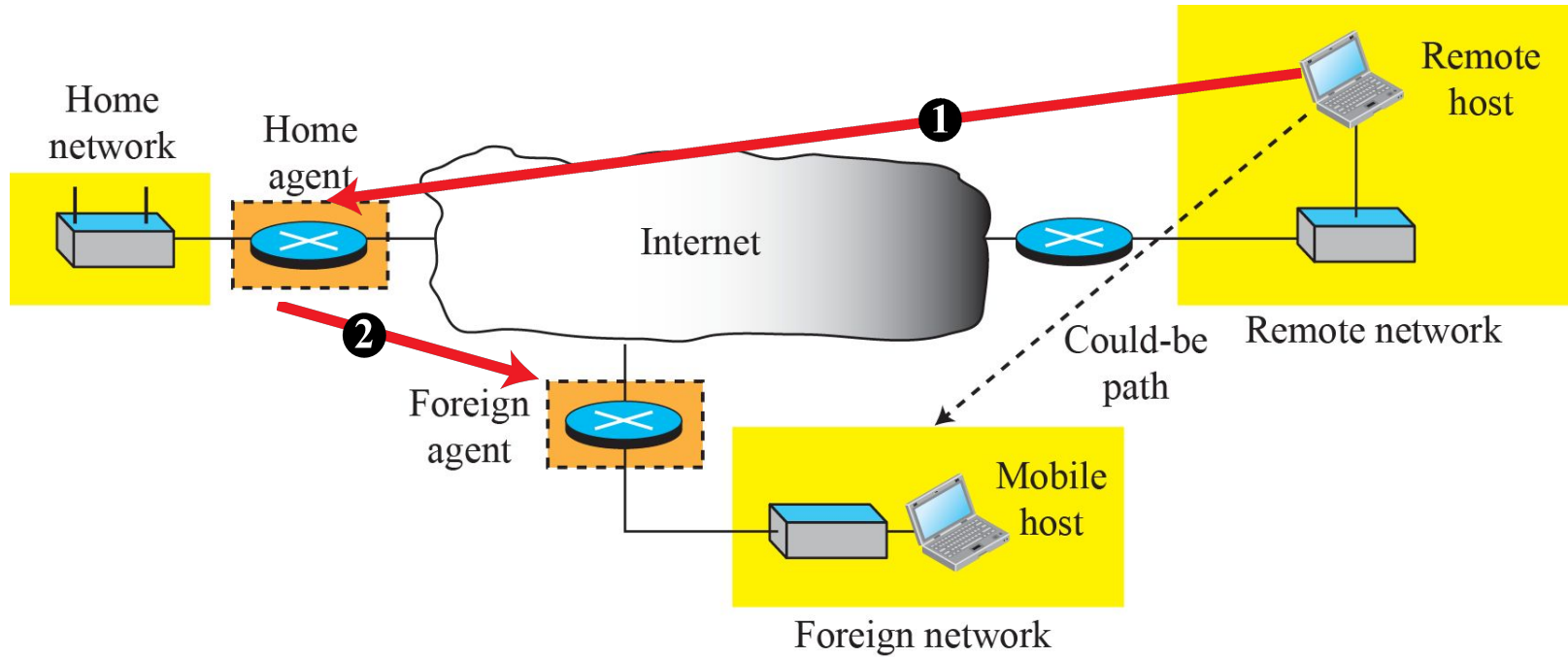
**□ *Triangle Routing***

**□ *Solution***

**Figure 6.59: Double crossing**



**Figure 6.60:** *Triangle routing*



# Chapter 6: Summary

- ☐ *Wireless LANs became formalized with the IEEE 802.11 standard, which defines two services: basic service set (BSS) and extended service set (ESS). The access method used in the distributed coordination function (DCF) MAC sublayer is CSMA/CA. The access method used in the point coordination function (PCF) MAC sublayer is polling.*
- ☐ *Bluetooth is a wireless LAN technology that connects devices (called gadgets) in a small area. A Bluetooth network is called a piconet.*
- ☐ *WiMAX is a wireless access network that may replace DSL and cable in the future.*

# Chapter 6: Summary (continued)

- ❑ *Cellular telephony provides communication between two devices. One or both may be mobile. A cellular service area is divided into cells. Cellular telephony has gone through four generations.*
- ❑ *A satellite network uses satellites to provide communication between any points on Earth. We have discussed several systems: including GEO, MEO, and LEO.*
- ❑ *Mobile IP is an enhanced version of the Internetworking Protocol (IP). A mobile host has a home address on its home network and a care-of address on its foreign network. When the mobile host is on a foreign network, a home agent relays messages (for the mobile host) to a foreign agent. A foreign agent sends relayed messages to a mobile host.*